Graphical symplectic algebra presents the dagger-compact PROP of affine Lagrangian relations

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Editorial note. Original author unresolved hyperlinks: the scalable axiom chart labels T.Id.1, T.Id.2, T.Id.3, T.One.1, T.One.2 and T.Z target a literal question-mark placeholder; T.Copy, T.Times and T.Zero target lem:tcopy, lem:ttimes and lem:tzero, which are absent from the complete native source. These nine author labels are displayed unchanged as plain text with no guessed link destination. Their graphical equations remain unchanged; original source and unresolved-link preamble are retained.

Allowed standard prerequisites (editorial).

- Strict monoidal/compact categories, affine subspaces, linear algebra over a field and row echelon forms. Native graphical/linear-algebra sections. The actual symplectic presentation and complete author normalization/uniqueness constructions are retained.

2 Graphical languages

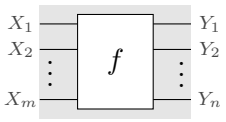
As mentioned in the previous section, symmetric monoidal categories give mathematical semantics for processes with a certain notion of sequential and parallel composition. A reference for symmetric monoidal categories can be found in Mac Lane [Lan71, §XI]. Throughout this section, let $(\mathsf{C}, \otimes, I)$ denote a symmetric monoidal category with tensor product $(-) \otimes (=) : \mathsf{C} \times \mathsf{C} \to \mathsf{C}$ and tensor unit $I \in \mathrm{Ob}(\mathsf{C})$. Denote the structure maps of the symmetric monoidal category as follows:

- **associators**: $\alpha_{X,Y,Z} : X \otimes (Y \otimes Z) \cong (X \otimes Y) \otimes Z$
- **unitors**: $\lambda_X : I \otimes X \cong X \qquad \textit{and} \qquad \rho_X : X \otimes I \cong X$
- **symmetries**: $\sigma_{X,Y} : X \otimes Y \cong Y \otimes X$

2.1 String diagrams

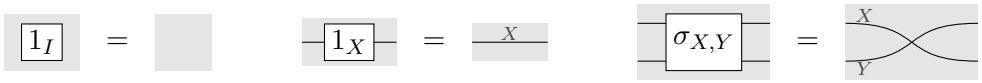
A symmetric monoidal category is *strict* when the unitors and associators are identities. Strict symmetric monoidal categories are very convenient to work with because they enjoy a graphical calculus in terms of *string diagrams*. An arrow $f : X_1 \otimes \cdots \otimes X_m \to Y_1 \otimes \cdots \otimes Y_n$

is drawn as a box with m input wires on the left, and n output wires on the right, where each wire is labeled by the corresponding object:



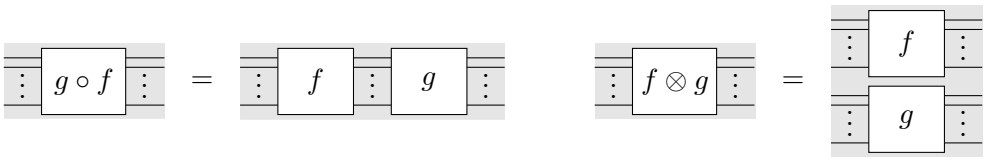
(1)

We will often drop the labels on wires when the objects can be inferred from context. The identity on the monoidal unit is drawn as an empty diagram, the identity on all other objects as a plain wire, and the components of the symmetry are drawn as wires crossing:



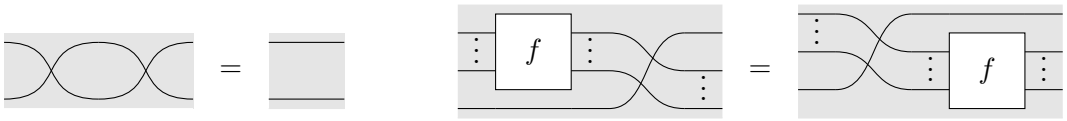
(2)

The composition of arrows is drawn by "plugging wires" and the monoidal product of arrows as "vertical juxtaposition":



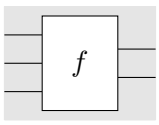
(3)

String diagrams for symmetric monoidal categories can be manipulated as one might expect:



(4)

Arrows of type $I \to X$ are called *states*, arrows of type $X \to I$ are called *effects*, and endomorphisms $I \to I$ are called *scalars*. A *prop* is a strict symmetric monoidal category which is generated by a single object under the tensor product. In other words, the monoid of objects under the tensor product is isomorphic the the natural numbers $\mathbb{N}$. Exploiting this isomorphism, we can always drop the wire labels because the types of arrows are uniquely identified by the number of input and output wires. For example, an arrow $f : 3 \to 2$ in a prop is represented as:



(5)

2.2 Strictification and scalable notation

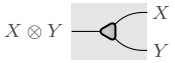
Without loss of generality, using the following well-known result, it is always possible to use string diagrams to reason about (not necessarily strict) symmetric monoidal categories:

Theorem 1. *Every symmetric monoidal category is symmetric monoidally equivalent to a strict symmetric monoidal category.*

This process is called *strictification*. There is an explicit strictification procedure for *monoidal* categories described by Wilson et al. [WGZ22] presented in terms of generators and relations. Similar results exist in the context of multiplicative linear logic: for example by Cockett and Seely [CS97]. Although the specific symmetric monoidal strictification procedure we use here has not explicitly appeared in the literature, it follows straightforwardly from the aforementioned results:

Definition 2. Given a symmetric monoidal category $(\mathsf{C}, \otimes, I)$, its *strictification* $([\mathsf{C}], +, [])$ has:

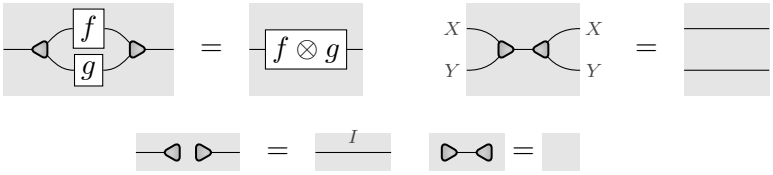
- **Objects:** in $\mathrm{List}(\mathrm{Ob}(\mathsf{C}))$.
- **Arrows:** For every arrow $f : X \to Y$ in C there is an induced arrow $[f] : [X] \to [Y]$ in $[\mathsf{C}]$. Moreover, for all objects X, Y in C we have the following arrows in $[\mathsf{C}]$:

  Dividers: 

  Gatherers: 

  We impose the following equations between arrows:

$$[f] \circ [g] = [f \circ g], \quad [1_X] = 1_{[X]} \tag{6}$$

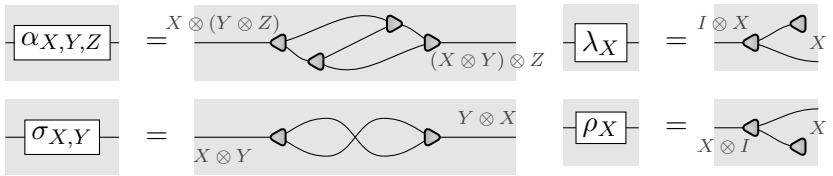


(7)

- **Strict monoidal structure:** The tensor unit is the empty list $[\,]$. The tensor product $+$ is list concatenation.
- **Symmetries:** given by permutations of list elements.

The *strictification functor* $[-] : (\mathsf{C}, \otimes, I) \simeq ([\mathsf{C}], +, [\,])$ takes arrows $f : X \to Y$ to $[f] : [X] \to [Y]$. For this reason we overload notation and take $[f] = f$. Equations (6) witness the functoriality and equations (7) witness the preservation of the symmetric monoidal structure. The dividers and gatherers are the components of the monoidal laxators for the strictification functor.

Remark 3. The components of the associator, unitors and symmetry in C are in are internalized to the following string diagrams in $[\mathsf{C}]$:

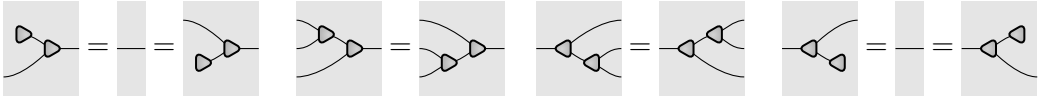


(8)

where their inverses are given by flipping the string diagrams around horizontally.

Scalable notation

In this paper, we only work with props. Even though props are already strict, it will still be useful to work with their strictifications. This allows for inductive arguments to be made using only string diagrams. To this end, first remark that in the strictification of a strict symmetric monoidal category, the laxators and oplaxators obey the following equations:



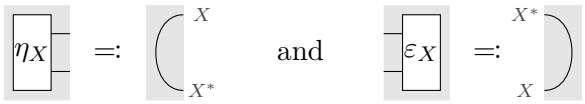
(9)

In order to make inductive arguments for props using only pictures, following Carette et al. [CHP19; Car20; CL22], as a matter of convention denote the object $[1]$ by a thin wire, and the objects $[n]$ for $n \in \mathbb{N}$ by thick wires (often labelled by their type n). For example, we use this notation to describe block matrices string digrammatically in subsection 3.3. Call this collection of graphical notation *scalable notation*.

2.3 Duality and flexsymmetry

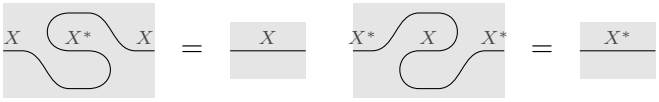
In this subsection, we extend symmetric monoidal categories with a notion of feedback:

Definition 4. A symmetric monoidal category is *compact-closed* when for every object X, there exists an object X^* together with a map $\eta_X : I \to X \otimes X^*$ called the *cup* and a map $\varepsilon_X : X^* \otimes X \to I$ called the *cap*, drawn as follows:



(10)

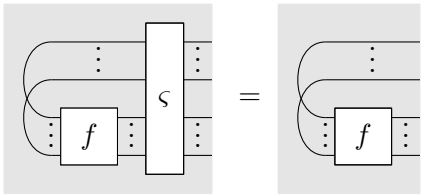
We require that $(X \otimes Y)^* = X^* \otimes Y^*$ and $I^* = I$. We also require the cups and caps to satisfy the following equations:



(11)

A *compact prop* is a compact-closed prop for which $1^* = 1$ and where cup and cap are symmetric, meaning that they satisfy $\varepsilon_1 \circ \sigma_{1,1} = \varepsilon_1$ and $\sigma_{1,1} \circ \eta_1 = \eta_1$.

Definition 5. Given a compact prop $(\mathsf{C}, +, 0)$, an arrow $f : m \to n$ is *flexsymmetric* if for any permutation ς of $m + n$ wires:



(12)

Notice that the cup and the cap are flexsymmetric. In some cases, compact props admit flexsymmetric presentations, that is, a presentation where all generators are flexsymmetric. In other words, the string diagrams can be treated as labeled graphs, and we no longer need to worry about the orientation of edges and arrows: only the connectivity of the diagram matters! This is very desirable from a computer science perspective, because it greatly simplifies the combinatorics of representing string diagrams with a data structure. More details on flexsymmetric presentations can be found in the PhD thesis of Carette [Car21].

Dagger structure

In many settings, a process $X \to Y$ is canonically associated to a process of the "reversed type" $Y \to X$. However, this map need not be the inverse. This idea can be captured as follows:

Definition 6. A *†-compact prop* is a compact prop $(\mathsf{C}, \otimes, I)$ equipped with a functor of props $(-)^\dagger : \mathsf{C}^{\mathsf{op}} \to \mathsf{C}$ such that:

- For all objects X in C, $X^\dagger = X$.
- For all maps $f : X \to Y$ in C, $(f^\dagger)^\dagger = f$.
- $\sigma_{1,1}^\dagger = \sigma_{1,1}^{-1}$ and $\eta_1^\dagger = \varepsilon_1$.

In such a setting, an arrow $f : X \to Y$ is an:

- *isometry* when $f^\dagger f = 1_X$.
- *coisometry* when $f f^\dagger = 1_Y$.
- *unitary* when it is both an isometry and coisometry (so that $f^\dagger = f^{-1}$).

3 Graphical linear and affine algebra

In this section we review string diagrammatic presentations of the props of affine and linear subspaces by Bonchi et al. [BSZ17; Bon+19], which we will respectively refer to as Graphical Linear Algebra (GLA) and Graphical Affine Algebra (GAA). GAA can be seen as a strict extension of GLA, thus we will first give its presentation and then restrict it to GLA. Note that the presentations which we give follow directly but differ slightly from those contained in the work of Bonchi et al. [BSZ17; Bon+19].

3.1 Props of linear and affine relations

Given some fixed set X, an X-relation $R : n \to m$ is a subset of the product $R \subseteq X^n \times X^m$. These form a compact prop:

Definition 7. Given a set X, the compact prop of *X-relations*, Rel_X, has:

- **Arrows** are relations between X^n and X^m.
- **Composition** of relations $R : n \to m$ and $S : m \to k$ is defined as:

$$S \circ R \coloneqq \{(x, z) \in X^n \times X^k \mid \exists y \in X^m : (x, y) \in R \wedge (y, z) \in S\} : n \to k$$

- **Identity** on n is given by the diagonal relation $\{(x, x) \in X^n \times X^n\} : n \to n$
- **Monoidal structure** is given by the object 0 and the Cartesian product:

$$S \times T \coloneqq \left\{ \left(\begin{bmatrix} x \\ y \end{bmatrix}, \begin{bmatrix} z \\ w \end{bmatrix} \right) \,\middle|\, (x, z) \in S \wedge (y, w) \in T \right\}$$

- **Symmetric structure** is generated by the relation:

$$\sigma_{1,1} \coloneqq \left\{ \left(\begin{bmatrix} x \\ y \end{bmatrix}, \begin{bmatrix} y \\ x \end{bmatrix} \right) \,\middle|\, x, y \in X \right\} : 2 \to 2$$

- **Compact structure** is generated by the relations:

$$\eta_1 \coloneqq \left\{ \left(\bullet, \begin{bmatrix} x \\ x \end{bmatrix} \right) \,\middle|\, x \in X \right\} : 0 \to 2 \quad \text{and} \quad \varepsilon_1 \coloneqq \left\{ \left(\begin{bmatrix} x \\ x \end{bmatrix}, \bullet \right) \,\middle|\, x \in X \right\} : 2 \to 0$$

Given a function $f : X^n \to X^m$ then its graph is a relation $\Gamma_f \coloneqq \{(x, f(x)) \mid x \in X\} : n \to m$. This assignment is functorial, so that the composition of functions corresponds to the relational composition of their graphs. We restrict this prop to the linear and affine settings. To this end, first recall some basic linear and affine algebra:

Definition 8. Given a field $\mathbb{K}$ and some $n \in \mathbb{N}$, a $\mathbb{K}$-*affine combination* of vectors $\vec{v}, \vec{w} \in \mathbb{K}^m$ is a vector $a \cdot \vec{v} + (1 - a) \cdot \vec{w}$ for some $a \in \mathbb{K}$. More generally, $\mathbb{K}$-*linear combination* of vectors $\vec{v}, \vec{w} \in \mathbb{K}^m$ is a vector $a \cdot \vec{v} + b \cdot \vec{w}$ for some $a, b \in \mathbb{K}$. A $\mathbb{K}$-*affine subspace* of $\mathbb{K}^n$ is subset $V \subseteq \mathbb{K}^n$ which is closed under affine combinations. Similarly, a $\mathbb{K}$-*linear subspace* of $\mathbb{K}^n$ is nonempty subset $V \subseteq \mathbb{K}^n$ which is closed under linear combinations.

Given any nonempty affine subspace $V \subseteq \mathbb{K}^n$, there exists a vector $\vec{a} \in \mathbb{K}^n$ such that the coset $V + \vec{a} \coloneqq \{\vec{v} + \vec{a} \mid \forall \vec{v} \in V\} \subseteq \mathbb{K}^n$ is a linear subspace. Call this vector, the *affine shift* associated to $V + \vec{a}$. Conversely, every linear subspace is an affine subspace, as affine combinations are particular linear combinations. Conceptually, affine subspaces are linear subspaces without a fixed origin. This justifies why we don't require affine subspaces to be inhabited: since there is no fixed origin, it need not contain the zero vector. Both of these notions of subspaces induce a compact prop:

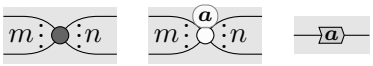
Definition 9. Given a field $\mathbb{K}$, the category, $\mathsf{AffRel}_{\mathbb{K}}$ (resp. $\mathsf{LinRel}_{\mathbb{K}}$), of $\mathbb{K}$-*affine (resp. linear) relations* is the sub-prop of $\mathsf{Rel}_{\mathbb{K}}$ whose arrows $n \to m$ are $\mathbb{K}$-affine (resp. linear) subspaces of the vector space $\mathbb{K}^n \times \mathbb{K}^m \cong \mathbb{K}^n \oplus \mathbb{K}^m$.

This is a category because affine (resp. linear) subspaces are closed under relational composition, and that the diagonal subset is a linear subspace. This is a compact prop because the cup, the cap and symmetry are also linear subspaces. By construction we have $\mathsf{LinRel}_{\mathbb{K}} \hookrightarrow \mathsf{AffRel}_{\mathbb{K}} \hookrightarrow \mathsf{Rel}_{\mathbb{K}}$. In $\mathsf{LinRel}_{\mathbb{K}}$ and $\mathsf{AffRel}_{\mathbb{K}}$, the arrows $0 \to n$ are exactly linear and affine subspaces of $\mathbb{K}^n$. In $\mathsf{LinRel}_{\mathbb{K}}$ there is a unique scalar $1_0 : 0 \to 0$, corresponding to the linear subspace $\mathbb{K}^0 \subset \mathbb{K}^0$. But in $\mathsf{AffRel}_{\mathbb{K}}$ we have two scalars, the identity and the empty affine subspace $\emptyset \subset \mathbb{K}^1$.

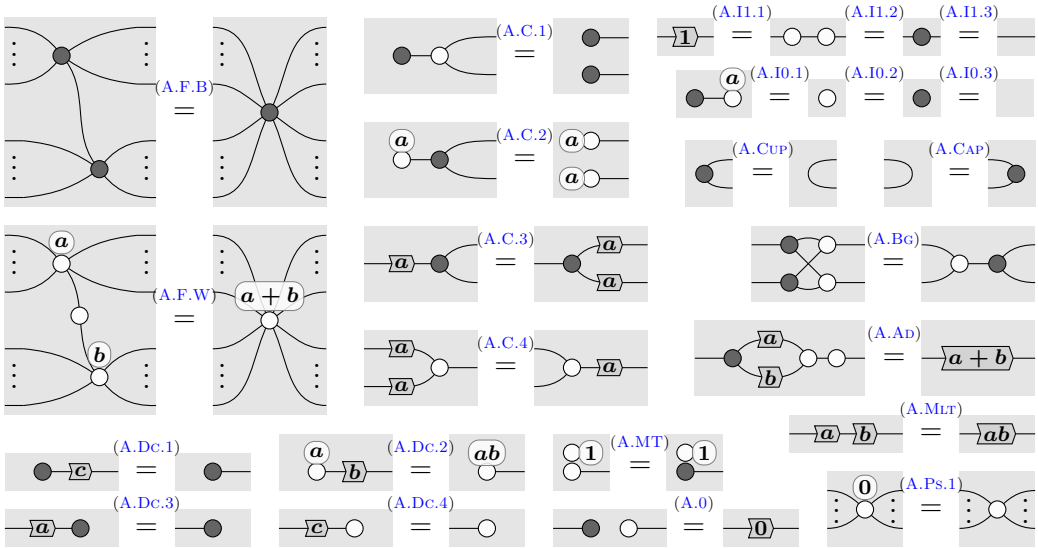
3.2 Graphical affine algebra and its interpretation

In this subsection, we review the graphical presentations for the props of affine and linear relations of Bonchi et al [Bon+19; BSZ17]. Let $\mathbb{K}$ be an arbitrary field, and $\mathbb{K}^*$ is the group of invertible elements of $\mathbb{K}$ under multiplication. We start with affine relations, and then restrict to linear relations:

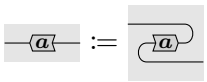
Definition 10. Let $\mathsf{GAA}_{\mathbb{K}}$ be the compact prop where the arrows are generated by *grey and white spiders*, as well as *scalar multiplication* for every $a \in \mathbb{K}$:



Modulo the equations making the grey and white spiders flexsymmetric as well as the equations, for all $a, b \in \mathbb{K}$ and $c \in \mathbb{K}^*$:



With the derived generators, for all $a \in \mathbb{K}$:



Theorem 11. *There is an isomorphism of compact props* $[\![-]\!]^{\mathsf{GAA}}_{\mathsf{AR}} : \mathsf{GAA}_{\mathbb{K}} \cong \mathsf{AffRel}_{\mathbb{K}}$

defined on generators, for all $n, m \in \mathbb{N}$ *and* $a \in \mathbb{K}$*:*

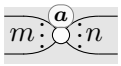
$$\left[\!\left[\; m \; \cdots \; n \; \right]\!\right]^{\mathsf{GAA}}_{\mathsf{AR}} := \left\{ \left(\begin{bmatrix} a \\ \vdots \\ a \end{bmatrix}, \begin{bmatrix} a \\ \vdots \\ a \end{bmatrix} \right) \in \mathbb{K}^n \oplus \mathbb{K}^m \;\middle|\; a \in \mathbb{K} \right\}$$

$$\left[\!\left[\; m \; \overset{a}{\circ} \; n \; \right]\!\right]^{\mathsf{GAA}}_{\mathsf{AR}} := \left\{ (\vec{b}, \vec{c}) \in \mathbb{K}^n \oplus \mathbb{K}^m \;\middle|\; \sum_{j=0}^{n-1} b_j + \sum_{k=0}^{m-1} c_k = a \right\}$$

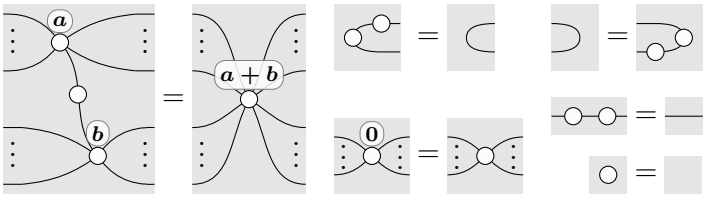
$$\left[\!\left[\; \boxed{a} \; \right]\!\right]^{\mathsf{GAA}}_{\mathsf{AR}} := \{(b, ab) \mid b \in \mathbb{K}\}$$

The notions of white and grey spiders labelled by a phase-group will occur throughout the rest of this paper in different incarnations:

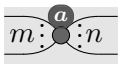
Definition 12. Given a group $\mathcal{G} = (G, +, 0)$, the compact prop of *$\mathcal{G}$-labelled white spiders* is generated by, for each $a \in G$ and $n, m \in \mathbb{N}$:



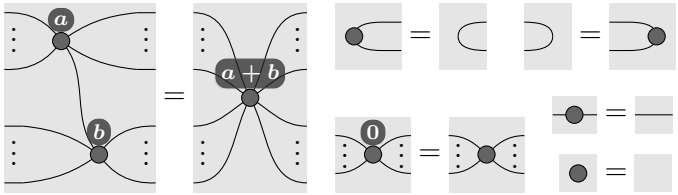
modulo flexsymmetry (see definition 5) and the following equations for all $a, b \in G$:



Similarly, the compact prop of *$\mathcal{G}$-labelled grey spiders* is generated by, for each $a \in G$:



modulo flexsymmetry and the following equations for all $a, b \in G$ and $n, m \in \mathbb{N}$:

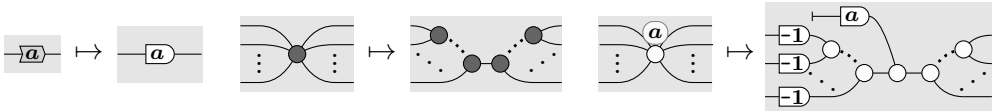


In both cases, say that $\mathcal{G}$ is the *phase group* of the white and grey spiders.

Therefore, we see that the white phase group of $\mathsf{GAA}_{\mathbb{K}}$ is the additive group of $\mathbb{K}$; whereas the grey phase group is the trivial group. As opposed to $\mathsf{AffRel}_{\mathbb{K}}$, we find that $\mathsf{LinRel}_{\mathbb{K}}$ has a more symmetric presentation:

Corollary 13. *By restricting the phase group of the grey spider in* $\mathsf{GAA}_{\mathbb{K}}$ *to be the trivial group, and removing the rules involving non-zero phases, we obtain a presentation* $\mathsf{GLA}_{\mathbb{K}}$ *for* $\mathsf{LinRel}_{\mathbb{K}}$*.*

Note that our presentations $\mathsf{GAA}_{\mathbb{K}}$ of $\mathsf{AffRel}_{\mathbb{K}}$ and $\mathsf{GLA}_{\mathbb{K}}$ of $\mathsf{LinRel}_{\mathbb{K}}$ differ from their original presentations by Bonchi et al. [BSZ17; Bon+19]. The original presentation of affine relations is recovered from ours via the translation:



Where the original presentation of $\mathsf{LinRel}_{\mathbb{K}}$ can be recoverd by restricting $a = 0$.

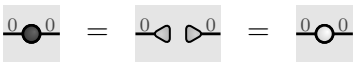
Remark 14. In our presentations $\mathsf{GAA}_{\mathbb{K}}$ and $\mathsf{GLA}_{\mathbb{K}}$ both grey and white spiders are flexsymmetric generators. In particular, this means that affine relations over $\mathbb{Q}$ and $\mathbb{F}_p$ for prime p admit flexsymmetric presentations because all scalars are generated by the scalars 1 and 0 under addition, multiplication and division.

This flexsymmetric presentation makes typesetting diagrams much easier, as one doesn't have to worry about the direction which wires are connected to generators; however, it will be useful to keep both presentations in mind, as the original non-flexsymmetric presentations are more natural from an algebraic perspective. For example, the concrete semantics of the constructions in the following subsection are more easily interpreted in the non-flexsymmetric presentation.

3.3 Scalable notation for graphical linear and affine algebra

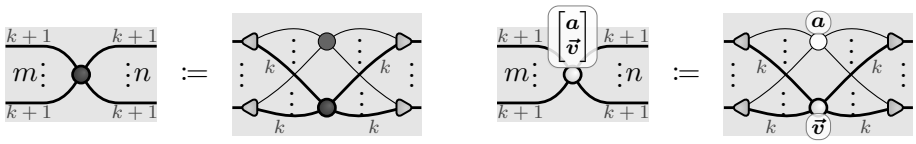
In this subsection we give notations for the strictification of $\mathsf{GAA}_{\mathbb{K}}$. This allows for many concepts in linear and affine algebra to be expressed concisely using pictures.

Definition 15. Scalable spiders are defined by induction on the colour of wire $k \in \mathbb{N}$. The base case is given by the identity on the tensor unit:



(13)

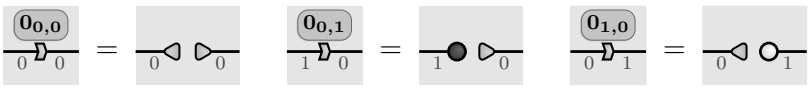
The inductive step is given by:



(14)

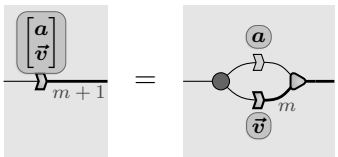
These scalable spiders satisfy analagous axioms as before, except now the white $[k]$-coloured spiders are indexed by the additive group of $\mathbb{K}^k$. As a matter of notation, for any $n, m \in \mathbb{N}$ take $\mathrm{Mat}_{\mathbb{K}}(n, m)$ to be the set of $m \times n$ matrices over $\mathbb{K}$.

Definition 16. Block matrices are defined by induction on the domain and codomain. The base cases are given by the unique matrices $0_{0,0} \in \mathrm{Mat}_{\mathbb{K}}(0,0)$, $0_{0,1} :\in \mathrm{Mat}_{\mathbb{K}}(1,0)$ and $0_{1,0} \in \mathrm{Mat}_{\mathbb{K}}(0,1)$:



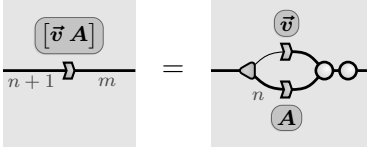
(15)

For the first inductive step given a vector $\vec{v} \in \mathbb{K}^m$, and an element $a \in \mathbb{K}$ we can build the matrix arrow for $[a\ \vec{v}^{\mathsf{T}}]^{\mathsf{T}} \in \mathbb{K}^{m+1}$:



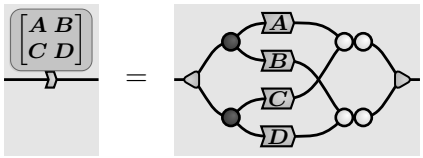
(16)

For the second inductive step, given a matrix $A \in \mathrm{Mat}_{\mathbb{K}}(n, m)$ and a vector $\vec{v} \in \mathbb{K}^m$, we can build the matrix arrow for $[\vec{v}\ A] \in \mathrm{Mat}_{\mathbb{K}}(n+1, m)$:



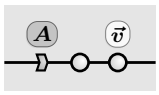
(17)

Unrolling this inductive definition, a block matrix divided once both horizontally and vertically is represented by:



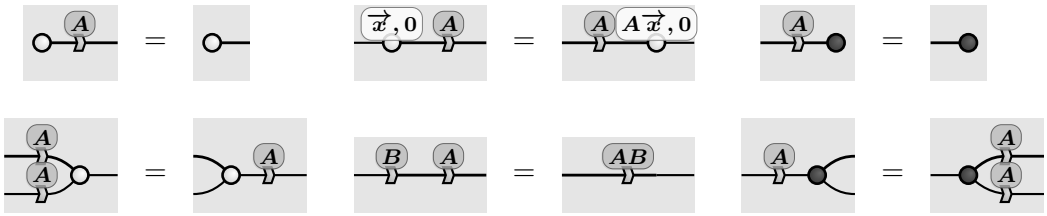
(18)

So that any affine transformation $(A, \vec{v})$; $x \mapsto Ax + \vec{v}$ is represented by the following diagram:



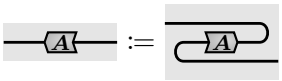
(19)

Note that the extra white nodes on the right hand sides of equations (17), (18) and (19) are artifacts of our flexsymmetric presentation. From this representation the following rules directly follow:



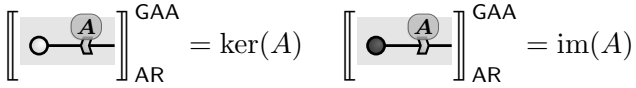
(20)

As a matter of notation, just like the converse of scalar multiplication, the converse of a matrix A is given by:



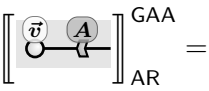
(21)

Using this notation, we can depict the image and the kernel of a matrix A using pictures:



(22)

And more generally the affine subspace of solutions for $\vec{v}$ in the equation $A\vec{v} = \vec{w}$:

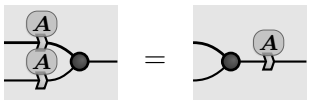


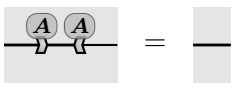
$= \{\vec{w} \mid A\vec{w} = \vec{v}\}$ (23)

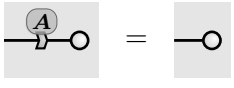
Note that in this relational setting, it is perfectly fine to ask for the the image or kernel of an affine *relation* by replacing the matrix arrows with arbitrary maps. We end this section by recalling diagrammatic reformulation of injectivity and surjectivity:

Proposition 17. *For any matrix A we have the following two separate columns of equivalent propositions:*

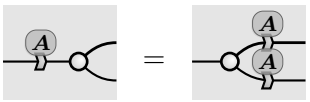
(1) *A is injective*

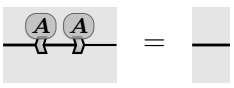
(2) 

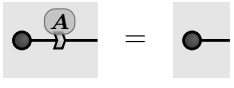
(3) 

(4) 

(1) *A is surjective*

(2) 

(3) 

(4) 

4 Graphical linear and affine Lagrangian algebra

In this section, we refine graphical treatments of linear and affine algebra to the symplectic setting. To this end, we first review some of the basic notions in linear symplectic geometry:

4.1 Symplectic linear and affine algebra

Definition 18. A *symplectic* $\mathbb{K}$*-vector space* is a $\mathbb{K}$-vector space V endowed with a bilinear map $\omega_V : V \oplus V \to \mathbb{K}$ called the *symplectic form* which is:

- *Alternating:* For all $\vec{v} \in V$, $\omega_V(\vec{v}, \vec{v}) = 0$.
- *Nondegenerate:* Given some $\vec{v} \in V$: if for all $\vec{u} \in V$ $\omega_V(\vec{v}, \vec{u}) = 0$, then $\vec{v} = \vec{0}$.

A *symplectic* $\mathbb{K}$*-linear map* between symplectic spaces $(V, \omega_V) \to (W, \omega_W)$ is a $\mathbb{K}$-linear map $S : V \to W$ that preserves the symplectic form, i.e. $\omega_W(S\vec{v}, S\vec{u}) = \omega_V(\vec{v}, \vec{u})$ for all $\vec{v}, \vec{u} \in V$. A *symplectomorphism* is a symplectic isomorphism. Similarly, a *symplectic* $\mathbb{K}$*-affine map* is a $\mathbb{K}$-affine map whose linear component is symplectic; and an affine symplectomorphism is an symplectic $\mathbb{K}$-affine isomorphism.

The vector space $\mathbb{K}^0$ has trivial symplectic structure. The next most simple example of a symplectic vector space is $\mathbb{K}^2$ equipped with the symplectic form

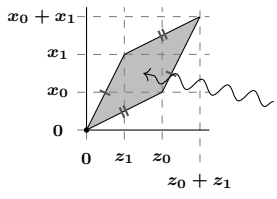
$$\omega_1\left(\begin{bmatrix} z_0 \\ x_0 \end{bmatrix}, \begin{bmatrix} z_1 \\ x_1 \end{bmatrix}\right) \coloneqq z_0 x_1 - x_0 z_1$$

Call the first component of $\mathbb{K}^2 \cong \mathbb{K} \oplus \mathbb{K}$ the Z-grading, and the second the X-grading. The vector space $\mathbb{K}^{2n}$ is also a symplectic vector space with the symplectic form:

$$\omega_n\left(\bigoplus_{j=0}^{n-1} \begin{bmatrix} z_j \\ x_j \end{bmatrix}, \bigoplus_{j=0}^{n-1} \begin{bmatrix} z'_j \\ x'_j \end{bmatrix}\right) \coloneqq \sum_{j=0}^{n-1} \omega_1\left(\begin{bmatrix} z_j \\ x_j \end{bmatrix}, \begin{bmatrix} z'_j \\ x'_j \end{bmatrix}\right)$$

Theorem 19 (Linear Darboux theorem)**.** *Every finite-dimensional symplectic vector space* (V, ω_V) *is symplectomorphic to the symplectic vector space* $(\mathbb{K}^{2n}, \omega_n)$*.*

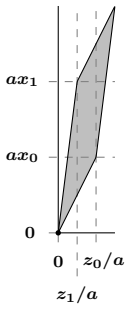
There is an intuitive geometric interpretation of ω_n: it refines the measure of oriented volumes in $\mathbb{K}^{2n}$, as $\bigwedge_{j=1}^{n} \omega_1$ is a volume form. For $n = 1$, ω_1 captures the area of a parallelogram given by points $(z_0, x_0), (z_1, x_1) \in \mathbb{K}^2$. This is visualized as follows, where the vertical axis denotes the X-component, and the horizontal axis denotes the Z-component:

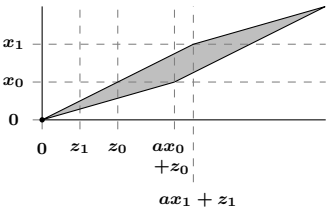


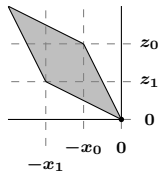
$$\omega_1\left(\begin{bmatrix} z_0 \\ x_0 \end{bmatrix}, \begin{bmatrix} z_1 \\ x_1 \end{bmatrix}\right) \tag{24}$$

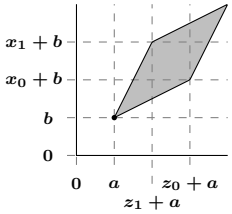
Therefore the linear and affine symplectomorphisms are the linear and affine transformations which preserve oriented volume between the Z and X-gradings:

Example 20. Using affine symplectomorphisms, the parallelogram from equation (24) can be:

- *Squeezed:* $\begin{bmatrix} a^{-1} & 0 \\ 0 & a \end{bmatrix} \mapsto$ 

- *Sheared:* $\begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix} \mapsto$ 

- *Rotated:* $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \mapsto$ 

- *Shifted:* $\left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} a \\ b \end{bmatrix} \right) \mapsto$ 

Notice that for $n = 1$ the symplectic form is a volume form and hence the correspondence is perfect, however, for $n \geqslant 2$, symplectomorphisms are still volume preserving but there are volume preserving maps that are not symplectomorphisms. Given a linear subspace S of a symplectic vector space (V, ω), its *symplectic complement* is the linear subspace:

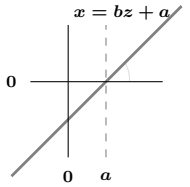
$$S^\omega \coloneqq \{\vec{v} \in V \mid \forall \vec{s} \in S : \omega(\vec{v}, \vec{s}) = 0\} \subseteq V$$

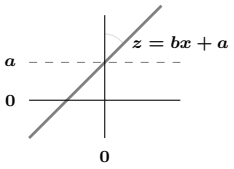
A linear subspace S of a symplectic vector space (V, ω) is:

- *isotropic* if $S \subseteq S^\omega$ (so that for all $\vec{s}, \vec{t} \in S$, $\omega(\vec{t}, \vec{s}) = 0$);
- *coisotropic* if $S^\omega \subseteq S$;
- *Lagrangian* if it is both isotropic and coisotropic (so that $S = S^\omega$).

Similarly, an affine subspace $(S, \vec{a})$ of a symplectic vector space (V, ω) is isotropic / coisotropic / Lagrangian when either S is isotropic / coisotropic / Lagrangian, or S is empty.

Example 21. Any line in the affine plane is given by an affine Lagrangian subspace of $(\mathbb{K}^2, \omega_1)$. These take of one of the two following forms, for some $a, b \in \mathbb{K}$:

$$\left\{ \begin{bmatrix} z \\ x \end{bmatrix} \in \mathbb{K}^2 \;\middle|\; x = bz + a \right\} \quad \text{visualized as} \quad$$ 

$$\left\{ \begin{bmatrix} z \\ x \end{bmatrix} \in \mathbb{K}^2 \;\middle|\; z = bx + a \right\} \quad \text{visualized as} \quad$$ 

Notice that the intersection of both classes of lines are the lines which pass through the origin and which are not completely vertical or horizontal. The entire plane, however, is the coisotropic subspace $\mathbb{K}^2 \subseteq \mathbb{K}^2$ whereas its orthogonal complement, the origin, is an isotropic subspace $\mathbb{K}^0 \subset \mathbb{K}^2$. An arbitrary single point on the plane is an affine coisotropic

subspace. From a geometric point of view, we see that the points and lines are affine isotropic subspaces of $\mathbb{K}^2$ because they have no volume; the line is affine Lagrangian subspace of $\mathbb{K}^2$ because it has the largest possible dimension without having any volume; and the entire plane is a coisotropic subspace because it is dual to the isotropic subspace given by the origin.

An *isotropic / coisotropic / Lagrangian, linear / affine relation* $m \to n$ is an isotropic / coisotropic / Lagrangian, linear / affine subspace of:

$$(\mathbb{K}^{2m} \oplus \mathbb{K}^{2n}, (\vec{v}, \vec{w}) \mapsto \omega_m(\vec{v}) - \omega_n(\vec{w}))$$

We must introduce this twisted symplectic form, as opposed to ω_{n+m}, so that these notions of "relations" are really stable under relational composition as described in definition 7:

Definition 22. The †-compact props of isotropic / coisotropic / Lagrangian, linear / affine relations, $\mathsf{IsotRel}_{\mathbb{K}}$, $\mathsf{CoIsotRel}_{\mathbb{K}}$, $\mathsf{LagRel}_{\mathbb{K}}$, $\mathsf{AffIsotRel}_{\mathbb{K}}$, $\mathsf{AffCoIsotRel}_{\mathbb{K}}$, $\mathsf{AffLagRel}_{\mathbb{K}}$ have:

- **Arrows** are given by isotropic / coisotropic / Lagrangian, linear / affine relations.
- **Composition** is given by relational composition.
- **Identity** on 1 is given by the diagonal on $\mathbb{K}^1$.
- **The monoidal structure** is given by the direct sum and 0.
- **The symmetry** on 1 is given by:

$$\left\{ \left(\begin{bmatrix} \vec{v} \\ \vec{w} \end{bmatrix}, \begin{bmatrix} \vec{w} \\ \vec{v} \end{bmatrix} \right) \,\middle|\, \vec{v}, \vec{w} \in \mathbb{K}^2 \right\} : 2 \to 2$$

- **The compact structure** on 1 is given by

$$\eta_1 \coloneqq \left\{ \left(\bullet, \begin{bmatrix} z \\ x \\ z \\ -x \end{bmatrix} \right) \,\middle|\, z, x \in \mathbb{K} \right\} : 0 \to 2 \qquad \varepsilon_1 \coloneqq \left\{ \left(\begin{bmatrix} z \\ x \\ z \\ -x \end{bmatrix}, \bullet \right) \,\middle|\, z, x \in \mathbb{K} \right\} : 2 \to 0$$

- **The dagger** of an arrow $R : n \to m$ is given by:

$$R^\dagger \coloneqq \left\{ \left(\bigoplus_{j=0}^{m-1} \begin{bmatrix} z_j \\ -x_j \end{bmatrix}, \bigoplus_{k=0}^{n-1} \begin{bmatrix} z'_k \\ -x'_k \end{bmatrix} \right) \,\middle|\, \left(\bigoplus_{k=0}^{n-1} \begin{bmatrix} z'_k \\ x'_k \end{bmatrix}, \bigoplus_{j=0}^{m-1} \begin{bmatrix} z_j \\ x_j \end{bmatrix} \right) \in R \right\} : m \to n$$

Our definition differs slightly from the way it is often stated in the literature, for example in the work of Weinstein [Wei09] and Baez et al. [BCR18]. This is because we have decided to work in the skeletal case, where all isomorphic objects are equal. However, both definitions are equivalent, so this difference is inconsequential. We must be careful as the relationship between the props $\mathsf{LinRel}_{\mathbb{K}}$ / $\mathsf{AffRel}_{\mathbb{K}}$ and their symplectic counterparts is subtle. First, for example, the symmetric monoidal functor $[\![-]\!]^{\mathsf{AR}}_{\mathsf{ALR}} : \mathsf{AffLagRel}_{\mathbb{K}} \to \mathsf{AffRel}_{\mathbb{K}}$ which forgets the symplectic structure is not a prop morphism, as it sends $n \mapsto 2n$. Furthermore the compact structure of $\mathsf{AffLagRel}_{\mathbb{K}}$ is different from the compact structure we defined on $\mathsf{AffRel}_{\mathbb{K}}$ because the latter is not Lagrangian. Remark that any (affine) symplectic map $R : (\mathbb{K}^{2n}, \omega_n) \to (\mathbb{K}^{2m}, \omega_m)$ induces an (affine) Lagrangian relation via its graph

$$\Gamma_R \coloneqq \{(\vec{v}, R\vec{v}) \mid \forall \vec{v} \in \mathbb{K}^{2n}\} : n \to m$$

which is functorial, so these categories of symplectic relations strictly generalize (affine) symplectic maps. It is helpful to get a geometric intuition for composition in these symplectic categories of relations:

Example 23. Consider the affine Lagrangian relation $R : 0 \to 1$ relating the point to the line $x = bz + a$; and consider another affine Lagrangian relation $S : 1 \to 0$ relating the line $x = b'z + a'$ to the point.

- If $b = b'$, so that both lines are parallel, but $a \neq a'$ so that they never meet, then $S \circ R$ is the empty set.
- On the other hand if $b = b'$ and $a = a'$, so that these are both the same lines then $S \circ R$ is the identity.
- Even if these two lines intersect only once, then $S \circ R$ is still the identity!

4.2 Generators and equations for affine Lagrangian relations

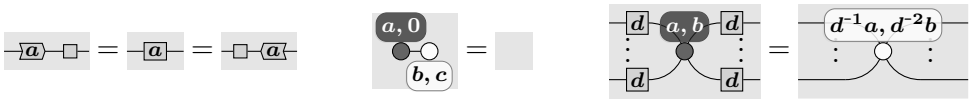
In order to give a graphical axiomatisation of $\mathsf{AffLagRel}_{\mathbb{K}}$, we extend the phase group of $\mathsf{GAA}_{\mathbb{K}}$ (recalling definition 12):

Definition 24. Let $\mathsf{GSA}_{\mathbb{K}}$ be the †-compact prop given by extending the phase groups of $\mathsf{GAA}_{\mathbb{K}}$; the grey spiders phase group extended along the group homorphism $\mathbb{K}^0 \hookrightarrow \mathbb{K}^2$ and the white spider along the group homorphism $\mathbb{K}^1 \hookrightarrow \mathbb{K}^2$:



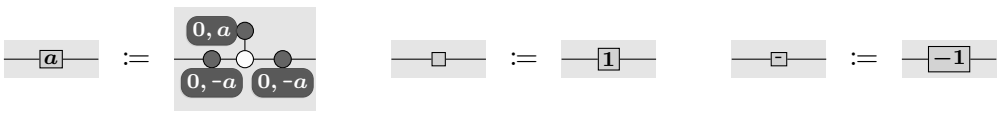
(25)

Modulo the following equations for all $a, b, c \in \mathbb{K}$ and $d \in \mathbb{K}^*$:



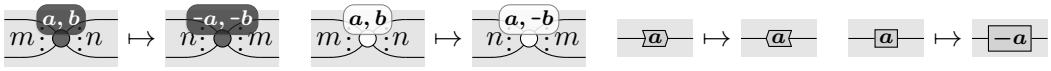
(26)

Where there are derived generators, for all $a \in \mathbb{K}$:



(27)

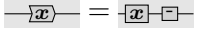
The †-structure is given by:



Call the first component of the phase groups for the grey and white spiders in equation (25) the *affine* phase, and the second component the *symplectic* phase.

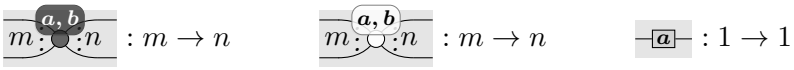
The derived generators in 27 are called *boxes*. The disinguished boxes with labels 1 and -1 are respectively the *symplectic Fourier transform* and *inverse symplectic Fourier transform*.

Recalling the geometric intuition from example 20, lifting scalar multiplication along the embedding $\mathsf{GAA}_{\mathbb{K}} \hookrightarrow \mathsf{GSA}_{\mathbb{K}}$ corresponds to squeezing. Similarly, the affine phase for the white spider corresponds to shifting the X-gradient. On the other hand, the affine phase for the grey spider corresponds to shifting the Z-gradient. The symplectic phases for the white and grey spiders corresponding to shearing in the X and Z direction. The symplectic Fourier transform corresponds to a $\pi/2$-rotation in the affine plane. Following equation (26), we can actually define multipliers from boxes:



From now on, we will take the boxes as primary, rather than multiplication by scalars. This is because:

Proposition 25. *The presentation* $\mathsf{GSA}_{\mathbb{K}}$ *of* $\mathsf{AffLagRel}_{\mathbb{K}}$ *with generators*



is flexsymmetric.

See appendix A.1 for an explicit presentation of $\mathsf{GSA}_{\mathbb{K}}$. Note that the symplectic setting is quite special: as discussed in remark 14 $\mathsf{AffRel}_{\mathbb{K}}$ and $\mathsf{LinRel}_{\mathbb{K}}$ do not obviously admit flexsymmetric presentations.

Definition 26. There is an interpretation $[\![-]\!]^{\mathsf{GSA}}_{\mathsf{ALR}} : \mathsf{GSA}_{\mathbb{K}} \to \mathsf{AffLagRel}_{\mathbb{K}}$ given by:

$$\left[\!\!\left[\;\begin{array}{c}\scriptstyle m\\ \scriptstyle a,b\\ \scriptstyle n\end{array}\;\right]\!\!\right]^{\mathsf{GSA}}_{\mathsf{ALR}} := \left\{ \left(\bigoplus_{j=0}^{m-1} \begin{bmatrix} z_j \\ x \end{bmatrix}, \bigoplus_{k=0}^{n-1} \begin{bmatrix} z'_k \\ x \end{bmatrix} \right) \middle| \begin{array}{l} \vec{z} \in \mathbb{K}^m, \vec{z'}; \in \mathbb{K}^n, x \in \mathbb{K} \quad \text{such that} \\ \displaystyle\sum_{j=0}^{m-1} z_j - \sum_{k=0}^{n-1} z'_k + bx = a \end{array} \right\}$$

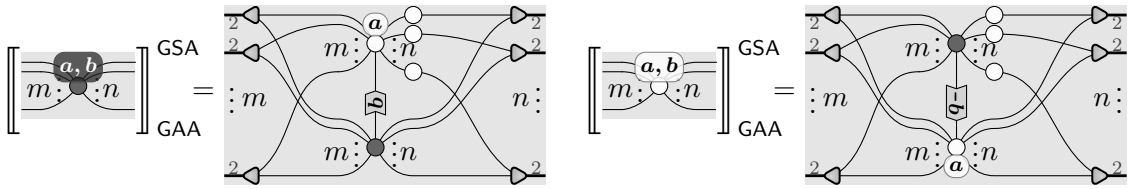
$$\left[\!\!\left[\;\begin{array}{c}\scriptstyle m\\ \scriptstyle a,b\\ \scriptstyle n\end{array}\;\right]\!\!\right]^{\mathsf{GSA}}_{\mathsf{ALR}} := \left\{ \left(\bigoplus_{j=0}^{m-1} \begin{bmatrix} z \\ x_j \end{bmatrix}, \bigoplus_{k=0}^{n-1} \begin{bmatrix} -z \\ x'_k \end{bmatrix} \right) \middle| \begin{array}{l} \vec{x} \in \mathbb{K}^m, \vec{x'} \in \mathbb{K}^n, z \in \mathbb{K} \quad \text{such that} \\ \displaystyle\sum_{j=0}^{m-1} x_j + \sum_{k=0}^{n-1} x'_k - bz = a \end{array} \right\}$$

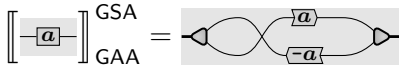
$$\left[\!\!\left[\;\boxed{0}\;\right]\!\!\right]^{\mathsf{GSA}}_{\mathsf{ALR}} := \left\{ \left(\begin{bmatrix} 0 \\ x \end{bmatrix}, \begin{bmatrix} 0 \\ x' \end{bmatrix} \right) \middle| x, x' \in \mathbb{K} \right\} \quad \left[\!\!\left[\;\boxed{c}\;\right]\!\!\right]^{\mathsf{GSA}}_{\mathsf{ALR}} := \left\{ \left(\begin{bmatrix} z \\ x \end{bmatrix}, \begin{bmatrix} -cx \\ z/c \end{bmatrix} \right) \middle| z, x \in \mathbb{K} \right\}$$

where $m, n \in \mathbb{N}$, $a, b \in \mathbb{K}$ and $c \in \mathbb{K}^*$.

Translating this to string diagrams, following Comfort and Kissinger [CK21], we have a factorization of $[\![-]\!]^{\mathsf{GSA}}_{\mathsf{ALR}} : \mathsf{GSA}_{\mathbb{K}} \to \mathsf{AffLagRel}_{\mathbb{K}}$ through $\mathsf{GAA}_{\mathbb{K}}$:

Definition 27. There is an interpretation $[\![-]\!]^{\mathsf{GSA}}_{\mathsf{GAA}} : \mathsf{GSA}_{\mathbb{K}} \to \mathsf{GAA}_{\mathbb{K}}$, defined on objects by $[\![m]\!]^{\mathsf{GSA}}_{\mathsf{GAA}} = 2m$ and on morphisms by:





Proposition 28. *The interpretations* $[\![-]\!]^{\mathsf{GSA}}_{\mathsf{ALR}} : \mathsf{GSA}_{\mathbb{K}} \to \mathsf{AffLagRel}_{\mathbb{K}}$ *and* $[\![-]\!]^{\mathsf{GSA}}_{\mathsf{GAA}} : \mathsf{GSA}_{\mathbb{K}} \to \mathsf{GAA}_{\mathbb{K}}$ *are †-symmetric monoidal functors, making the following diagram of †-compact closed categories commute:*

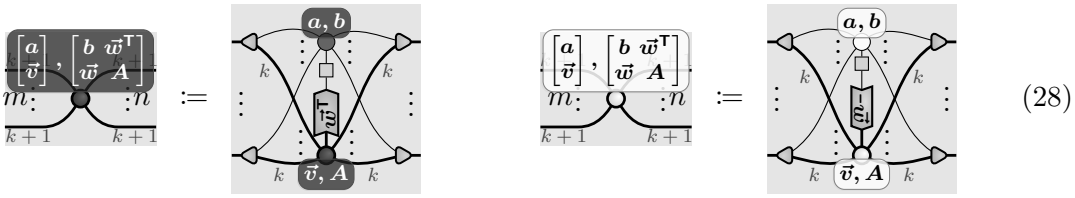
$$\begin{array}{ccc} \mathsf{GSA}_{\mathbb{K}} & \xrightarrow{[\![-]\!]^{\mathsf{GSA}}_{\mathsf{GAA}}} & \mathsf{GAA}_{\mathbb{K}} \\ {\scriptstyle [\![-]\!]^{\mathsf{GSA}}_{\mathsf{ALR}}}\downarrow & & \downarrow{\scriptstyle [\![-]\!]^{\mathsf{GAA}}_{\mathsf{AR}}} \\ \mathsf{AffLagRel}_{\mathbb{K}} & \xrightarrow[[\![-]\!]^{\mathsf{ALR}}_{\mathsf{AR}}]{} & \mathsf{AffRel}_{\mathbb{K}} \end{array}$$

Where the vertical arrows are morphisms of †-compact props.

4.3 Scalable notation for graphical symplectic algebra

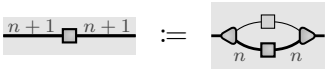
In this subsection we extend the scalable notation for affine relations introduced in section 3.3 to the setting of affine Lagrangian relations. First, note that the the matrix arrows in $\mathsf{GAA}_{\mathbb{K}}$ lift along the embedding $\mathsf{GAA}_{\mathbb{K}} \hookrightarrow \mathsf{GSA}_{\mathbb{K}}$ given in definition 24. However, now the $[n]$-coloured grey and white spiders both have phase groups $\mathbb{K}^n \times \mathrm{Sym}_n(\mathbb{K})$ (recalling definition 12), where $\mathrm{Sym}_n(\mathbb{K})$ denotes the Abelian group of symmetric $n \times n$ matrices over $\mathbb{K}$ under addition.

Definition 29. Define scalable grey and white spiders in $\mathsf{GSA}_{\mathbb{K}}$ by induction on the colour of wires $k \in \mathbb{N}$. The base case is trivial. For the inductive step, take $n, m \in \mathbb{N}$, $a, b \in \mathbb{K}$, $\vec{v}, \vec{w} \in \mathbb{K}^k$ and $A \in \mathrm{Sym}_k(\mathbb{K})$, we define $[k+1]$-coloured spiders as:



(28)

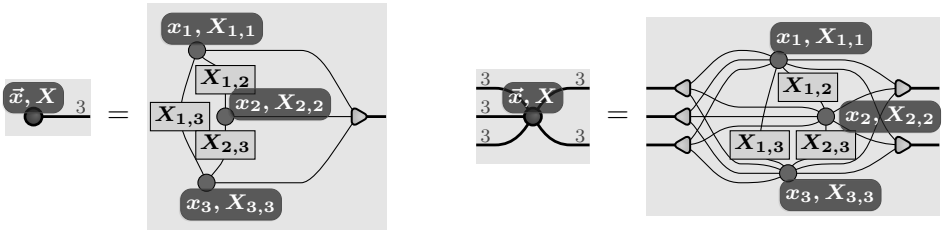
where the scaled box is also inductively defined. The base case is trivial, and the inductive step is given by:



(29)

A k-coloured grey spider with n inputs and m outputs parametrizes an undirected open graph with edges coloured by $\mathbb{K}$, vertices coloured by $\mathbb{K}^2$, with n distinguished inputs and m distinguished outputs.

Example 30. On the left hand side we have a graph *state* with $n = 0$, $m = 1$ and $k = 3$. On the right hand side, we have a spider with $m = 3$, $n = 2$ and $k = 3$:



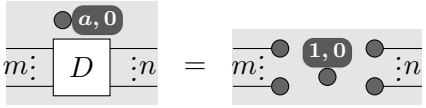
(30)

The axioms of $\mathsf{GSA}_{\mathbb{K}}$ admit very natural scalable generalizations which we have included in appendix A.3. This notation is very powerful, and makes the proof of completeness much more understandable and elegant.

4.4 Completeness and normal form

In this subsection, we show that the interpretation $[\![-]\!]^{\mathsf{GSA}}_{\mathsf{ALR}} : \mathsf{GSA}_{\mathbb{K}} \cong \mathsf{AffLagRel}_{\mathbb{K}}$ is an isomorphism of †-compact props. We have already proven that it is a †-compact functor in proposition 28. Fullness follows immediately from the result of Comfort and Kissinger [CK21, Lemma 4.6]. Therefore it remains to show that it is faithful. To prove this, we establish a normal form. If a diagram has empty semantics, then a simple induction argument yields the following unique normal form:

Proposition 31. *For any $m, n \in \mathbb{N}$ and $D \in \mathsf{GSA}_{\mathbb{K}}(m, n)$ and $a \in \mathbb{K}^*$:*



The normal form for diagrams with nonempty semantics takes more work. To this end, we further expose the graph theoretical properties of $\mathsf{GSA}_{\mathbb{K}}$. Consider a subclass of diagrams, which is inspired by that of Duncan et al. [Dun+20] in the context of the qubit ZX-calculus:

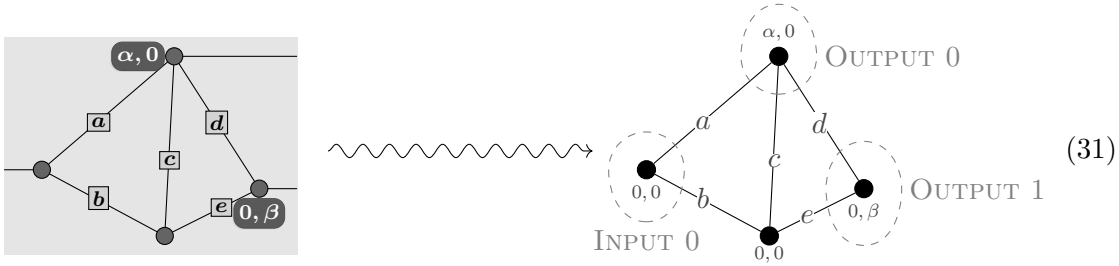
Definition 32. A $\mathsf{GSA}_{\mathbb{K}}$-diagram is *graph-like* when:

- All spiders are grey spiders.
- Grey spiders are only connected via edges containing a box.
- There are no self-loops.
- Every input or output is connected to a grey spider.
- Every grey spider is connected to at most one input or output.

Graph like diagrams induce undirected graph with edges coloured by $\mathbb{K}$, vertices coloured by $\mathbb{K}^2$, with distinguished vertices corresponding to spiders which are connected to specific inputs and outputs. Given a graph-like diagram:

- To each (a, b)-phased grey spider, associate an (a, b)-labelled vertex.
- To each wire between two spiders mediated by an a-labelled box, associate an a-labelled edge between the corresponding internal vertices.
- To each spider connected connected to an input/output, distinguish the corresponding edge as being associated to the specific input/output. Call such a vertex a *boundary vertex*. Call vertices not associated to inputs/outpus *internal*.

Example 33. The following $\mathsf{GSA}_{\mathbb{K}}$-diagram is graph-like:

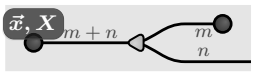


(31)

where the top three vertices are boundary vertices, and the bottom one is internal.

Proposition 34. *Every* $\mathsf{GSA}_{\mathbb{K}}$*-diagram can be rewritten to graph-like form.*

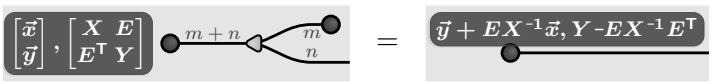
Because $\mathsf{GSA}_{\mathbb{K}}$ is compact closed, we can bend the input wires of diagrams into output wires. As long as we remember which outputs originally came from bending inputs, we only need to work with *states* $0 \to n$. Graph-like states take the following simple form in the scalable notation, for some $m, n \in \mathbb{N}$, $\vec{x} \in \mathbb{K}^{m+n}$ and $X \in \mathrm{Sym}_{m+n}(\mathbb{K})$:



(32)

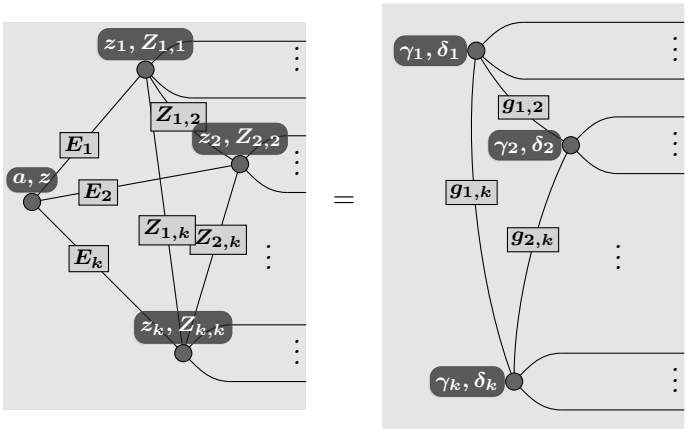
This graph-like state has m internal vertices, and n outputs. As before, X encodes the adjacency matrix of the underlying weighted graph with $m + n$ vertices (compare with example 30 and equations (31) and (32)). It is possible to simplify graph-like diagrams by reducing the number of internal vertices using the following result.

Proposition 35. *For any* invertible $X \in \mathrm{Sym}_m(\mathbb{K})$, $Y \in \mathrm{Sym}_n(\mathbb{K})$, $E \in \mathrm{Mat}_{\mathbb{K}}(n, m)$, $\vec{x} \in \mathbb{K}^m$, *and* $\vec{y} \in \mathbb{K}^n$*:*



This proposition encompasses both *local complementation* and *pivoting* of graphs in Bouchet [Bou94], generalised to coloured graphs by Kante and Rao [KR13]. This has been used in the context of the stabiliser fragment of the ZX-calculus, for example by Backens [Bac14], Duncan et al. [Dun+20], Booth and Carette [BC22], and Poór et al. [Poó+23]:

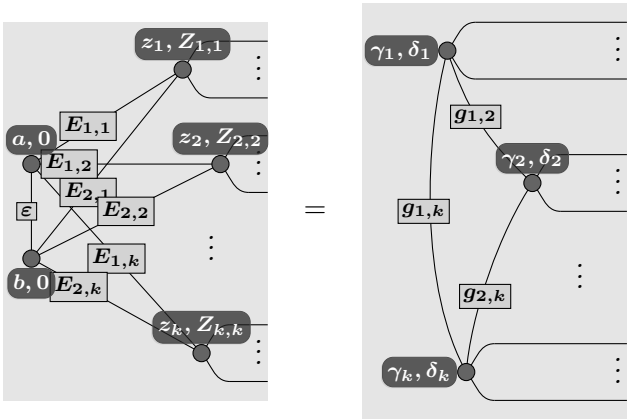
Proposition 36 (Local complementation)**.** *For any* $z \in \mathbb{K}^*$, $a \in \mathbb{K}$, $\vec{z} \in \mathbb{K}^k$, $E \in \mathrm{Mat}_{\mathbb{K}}(1,k)$, $Z \in \mathrm{Sym}_k(\mathbb{K})$ *where*



Where $\gamma_i \coloneqq z_i + E_i a z^{-1}$, $\delta_i \coloneqq Z_{i,i} - z^{-1}E_i^2$, *and* $g_{i,j} \coloneqq Z_{i,j} - z^{-1}E_iE_j$.

The new edge weights $g_{i,j}$ in the right-hand side are obtained from the edge weights E_{i_j} of the left-hand side by performing a local complementation (in the sense of Kante and Rao [KR13]) around the vertex 1, which is then deleted.

Proposition 37 (Pivoting)**.** *For any* $\varepsilon \in \mathbb{K}^*$, $a, b \in \mathbb{K}$, $\vec{z} \in \mathbb{K}^k$, $E \in \mathrm{Mat}_{\mathbb{K}}(2,n)$, $Z \in \mathrm{Sym}_k(\mathbb{K})$ *where*



Where

$\gamma_i \coloneqq z_i + \varepsilon^{-1}(aE_{2,i} + bE_{1,i})$, $\delta_i \coloneqq Z_{i,i} - 2\varepsilon^{-1}E_{1,i}E_{2,i}$, *and* $g_{i,j} \coloneqq -\varepsilon^{-1}(E_{1,i}E_{2,j} + E_{1,j}E_{2,i})$.

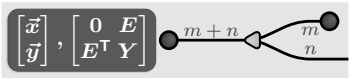
The new edge weights $g_{i,j}$ in the right-hand side are obtained from the edge weights E_{i_j} of the left-hand side by performing a pivot (in the sense of Kante and Rao [KR13]) around the edge weighted by ε. The two vertices connected by this edge are then deleted. Using proposition 36, it is clear that we can eliminate any internal spider with phase (x, z) where $z \neq 0$. Having done this, we then use proposition 37 to eliminate any remaining internal spiders that are connected. Therefore, we can refine the notion of graph-like diagram:

Definition 38. A graph-like diagram is in *Affine with Phases form* (AP-form) when:

- There are no inputs;
- The internal spiders have phases of the form $(x, 0)$ for $x \in \mathbb{K}$;
- Internal spiders are only connected to boundary spiders.

Proposition 39. *Every graph-like diagram $0 \to n$ can be put into AP-form.*

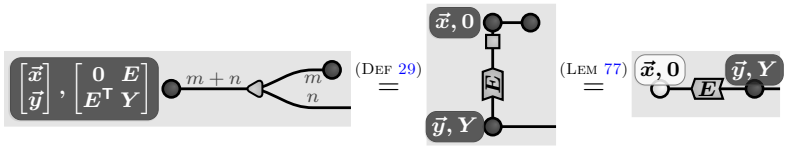
Remark 40. In the scalable notation, the AP-form can be seen to be the following refinement of the graph-like form which was presented in equation (32):



(33)

for $m, n \in \mathbb{N}$, $\vec{x} \in \mathbb{K}^m$, $\vec{y} \in \mathbb{K}^n$, $E \in \mathrm{Mat}_{\mathbb{K}}(n, m)$, and $Y \in \mathrm{Sym}_n(\mathbb{K})$. Here there are m internal spiders with phases $(x_k, 0)$, connected to n boundary spiders via the matrix E.

The AP-form also appeared in the completeness result of odd-prime-dimensional stabiliser ZX-calculus of Poór et al. [Poó+23]. Our scalable presentation of the AP-formal form can be easily rewritten to more closely resemble their presentation:



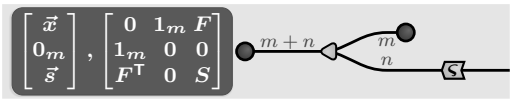
(34)

The history of this form is rather complicated, and we will discuss the attribution and history at the end of this section. The AP-form, can be row-reduced, which leads to the following refinement:

Definition 41. A diagram in AP-form defined by $(E, Y, \vec{x}, \vec{y})$ is in *reduced AP-form* if its semantics are empty, or it satisfies the following:

- E is in reduced row echelon form;
- E is surjective (none of the rows of E are zero);
- $\mathrm{im}(E^\mathsf{T}) \subseteq \ker(Y)$ (if the jth column of E is a pivot, then the jth column of Y is 0_n);
- $\mathrm{im}(E^\mathsf{T}) \subseteq \ker(\vec{y}^\mathsf{T})$ (if the jth column of E is a pivot, then the jth entry of $\vec{y}$ is 0).

Remark 42. In the scalable notation, the reduced AP-form can be stated graphically, for some $m \leqslant n \in \mathbb{N}$, $\vec{x} \in \mathbb{K}^m$, $\vec{s} \in \mathbb{K}^{n-m}$, $F \in \mathrm{Mat}_{\mathbb{K}}(n-m, m)$ and $S \in \mathrm{Sym}_{n-m}(\mathbb{K})$, up to a permutation matrix $\varsigma \in \mathrm{Mat}_{\mathbb{K}}(n, n)$:



(35)

Note that the permutation ς allows for the pivots 1_m to be separated into a different block than F, so that $E = \begin{bmatrix}1_m & F\end{bmatrix}\varsigma$, and similarly for $\vec{s}$ and 0_m, so that $y = \varsigma^\mathsf{T}\begin{bmatrix}0_m^\mathsf{T} & \vec{s}^\mathsf{T}\end{bmatrix}^\mathsf{T}$. Therefore, pulling the permutation out to the right makes the normal form easier to state diagrammatically.

Proposition 43. *Every $\mathsf{GSA}_{\mathbb{K}}$-diagram can be put into reduced AP-form.*

The reduced AP-form is unique:

Proposition 44. *For any nonempty affine Lagrangian state there is exactly one equivalent diagram in reduced-AP form.*

Recall that $[\![-]\!]^{\mathsf{GSA}}_{\mathsf{ALR}} : \mathsf{GSA}_{\mathbb{K}} \to \mathsf{AffLagRel}_{\mathbb{K}}$ is full. Moreover, it faithful on diagrams with empty semantics by proposition 31. Because, in addition, every nonempty $\mathsf{GSA}_{\mathbb{K}}$ diagram can be rewritten to the reduced-AP form by proposition 43, which is unique by proposition 44, we have that:

Theorem 45. *The interpretation* $[\![-]\!]^{\mathsf{GSA}}_{\mathsf{ALR}} : \mathsf{GSA}_{\mathbb{K}} \cong \mathsf{AffLagRel}_{\mathbb{K}}$ *is a †-compact prop isomorphism.*

This also gives a complete representation for Lagrangian relations as an immediate corollary:

Corollary 46. *There is an isomorphism of †-compact props* $[\![-]\!]^{\mathsf{GSA}^0}_{\mathsf{ALR}} : \mathsf{GSA}^0_{\mathbb{K}} \cong \mathsf{LagRel}_{\mathbb{K}}$*, where* $\mathsf{GSA}^0_{\mathbb{K}}$ *obtained by restricting the affine phases of the white and grey spiders in* $\mathsf{GSA}_{\mathbb{K}}$ *to be 0 and dropping the axioms involving nonzero affine phase.*

See appendix A.2 for an explicit presentation of $\mathsf{GSA}^0_{\mathbb{K}}$.

A note on the attribution of the (reduced) AP-form

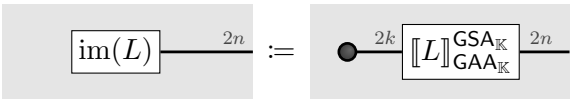
What we have called the "reduced AP-form" has a complicated history, originating from within the field of quantum information theory, which we review here in order to give correct attribution. A closely related form, called the "standard form" for qubit stabiliser codes seems to first appear in the work of Cleve and Gottesman [CG97]. When the stabiliser code is taken to have maximal dimension, the standard form for the stabiliser tableau is essentially the reduced AP-form for a linear Lagrangian subspace over the field $\mathbb{F}_2$. The connection of this form for qubit stabiliser states to graphs was developed by Elliott et al. [EEC08]; they decorate the graphs to recover the "phase-information" which is lost in the representation of a stabiliser state via its standard form. Hu and Khesin [HK22] refined the results of Elliott et al. [EEC08], providing a unique graph-like representation of qubit stabiliser states which also accounted for phase information, in terms of its affine support and a phase-polynomial. Note that the work of Elliott et al. [EEC08] was itself is directly inspired by that of Van den Nest et al. [VDD04], where they showed that, modulo scalars, qubit stabiliser states can be represented, non-uniquely by "quantum graph states" up to the action of the local Clifford operators. In the ZX-calculus community, Backens [Bac14] used Van den Nest et al.'s pseudonormal form of quantum graph states in order to give the first complete presentation of the qubit stabiliser ZX-calculus. Later on, McElvaney and Backens reformulated the unique normal form of Hu and Khesin [HK22] in the context of the qubit ZX-calculus [MB23]. Because of the connection which they draw to phase-polynomials, they named this incarnation of the "canonical form" with phase information the "phase-polynomial form" for a qubit stabiliser ZX-diagram, analagous to our AP-from. They further refine this form into the unique "canonical phase-polynomial form" for qubit stabiliser ZX-diagrams. This canonical phase-polynomial form is analagous to our reduced AP-form. To make the connection between the "canonical phase-polynomial form" of McElvanney and Backens [MB23] and the AP-form as described in our paper, we must take a detour through odd-prime-dimensional quantum circuits. In the setting of odd-prime-dimensional quantum systems, Bahramgiri and Beigi [BB07b; BB07a] had already noted that local complementation of $\mathbb{F}_p$-coloured graphs, as defined in the work of Kante and Rao [KR13], provides a suitable generalisation of the results of Van den Nest et al. [VDD04]. Booth and Carette [BC22] used this to prove completeness of a ZX-calculus for odd-prime-dimensional stabiliser quantum systems, using a proof method strongly inspired by [Bac14]. On the other hand, inspired by the phase-polynomial form and the unique canonical phase-polynomial form for qubit stabiliser states of McElvanney and Backens [MB23], Poór et al. [Poó+23] adapted these forms to the odd-prime-dimensional

qudit stabiliser fragment of the ZX-calculus. They dubbed the qudit analogue of the canonical phase-polynomial form the "affine with phases form" (AP-form), and the analogue of the canonical phase-polynomial form the "reduced affine with phases form" (reduced AP-form). By contrast to the quantum graph-state approach, the uniqueness of the reduced AP-form provided a cleaner completeness proof for the odd-prime-dimensional qudit stabiliser ZX-calculus. Since Comfort and Kissinger [CK21] proved that $\mathsf{AffLagRel}_{\mathbb{F}_p}$ is a projective representation for the odd-prime-dimensional qudit stabiliser ZX-calculus the year before (see section 6.2.1), our generalization of this normal form to $\mathsf{AffLagRel}_{\mathbb{K}}$ was a natural next-step. Although, annoyingly. instead of representing qubit stabiliser circuits, $\mathsf{AffLagRel}_{\mathbb{F}_2}$ is isomorphic to the prop of "pure circuits in Spekken's toy model" which was axiomatized separately by Backens and Duman [BD15]; hence the difficulty in giving correct attribution. In fact, Cockett et al. [CKP22] had already remarked that any affine Lagrangian relation can be represented by a generator matrix in a way which is analogous to the canonical form for a stabiliser state. When translated into string diagrams this coincides with the reduced AP-form we have given here; however their statement is purely semantic, and does not concern the completeness of a graphical language.

5 Graphical linear and affine coisotropic algebra

In this section, we extend the presentation $\mathsf{GSA}_{\mathbb{K}}$ of $\mathsf{AffLagRel}_{\mathbb{K}}$ to one for $\mathsf{AffCoIsotRel}_{\mathbb{K}}$. We also obtain a presentation of $\mathsf{CoIsotRel}_{\mathbb{K}} \cong \mathsf{IsotRel}_{\mathbb{K}}$ as an immediate corollary. First, recall from subsection 3.3 that we can not only take the images of matrices, but we can also take images of affine (resp. linear) relations. Because affine (resp. linear) Lagrangian relations are structure preserving affine (resp. linear) relations, we can also ask for their images:

Definition 47. Given an affine Lagrangian relation $L : k \to n$, the *image of* L is the image of the underlying affine relation $2k \to 2n$:



And likewise, for the image of linear Lagrangian relations.

Recall the following result of Comfort [Com23, Theorem 4.2]:

Lemma 48. *The set of nonempty affine (resp. linear) coisotropic subspaces of a symplectic vector space* $(\mathbb{K}^{2n}, \omega_n)$ *of dimension* $n + k$ *are in bijection with the set of the images of isometries of type* $k \to n$ *in* $\mathsf{AffLagRel}_{\mathbb{K}}$ *(resp.* $\mathsf{LagRel}_{\mathbb{K}}$*).*

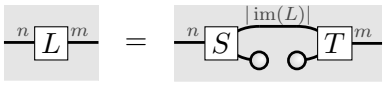
Therefore, we only need to add a single generator of type $0 \to 1$ to $\mathsf{GSA}_{\mathbb{K}}$ (resp. $\mathsf{GSA}^0_{\mathbb{K}}$) to get a presentation for $\mathsf{AffCoIsotRel}_{\mathbb{K}}$ (resp. $\mathsf{CoIsotRel}_{\mathbb{K}}$). Geometrically, this corresponds to the linear relation which relates the point to affine plane. In fact, it will be slightly more convenient, but equivalent, to instead add its converse: the relation which discards the entire plane. We now must find which additional equations this generator satisfies. To this end, we investigate more of the the structure of $\mathsf{AffLagRel}_{\mathbb{K}}$ and $\mathsf{LagRel}_{\mathbb{K}}$. First, we observe that there is an alternative presentation of affine/linear coisotropic subspaces due to Comfort [Com23, Theorem 4.5]:

Lemma 49. *Given* $L : k \to n$ *and* $K : m \to n$ *in* $\mathsf{AffLagRel}_{\mathbb{K}}$ *(or in* $\mathsf{LagRel}_{\mathbb{K}}$*), then:*

$$\mathrm{im}(L) = \mathrm{im}(K) \iff LL^\dagger = KK^\dagger$$

Second, observe that linear and affine Lagrangian relations have the following bipartite decomposition:

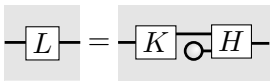
Lemma 50. *Any affine (resp. linear) Lagrangian relation $L : n \to m$ can be decomposed in the form:*



where $S : n \to n$ and $T : m \to m$ are affine (resp. linear) symplectomorphisms.

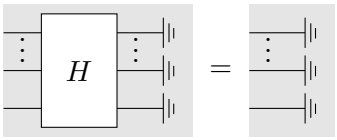
Note that affine symplectomorphisms are precisely the isomorphisms which are precisely the unitaries in $\mathsf{AffLagRel}_{\mathbb{K}}$, and similarly, for the linear symplectomorphisms in $\mathsf{LagRel}_{\mathbb{K}}$. It seems that this decomposition is well-known, where it is related to "coisotropic reduction" as in Weinstein [Wei09, section 3.1]. However, we arrived at this decomposition from a slightly different angle than it usually stated in the literature. For this reason, it is evident that this decomposition is an instance of the "generalized compact singular value decomposition" in the context of dagger categories of Cockett and Lemay [CL23, definition 3.5]. So in some sense, this a symplectic version of the singular value decomposition of complex matrices. Because of this similarity to the singular value decomposition, we have a symplectic version of the "essential uniqueness of purification" of mixed quantum circuits:

Lemma 51. *Given (affine) Lagrangian relations $L : n \to a$ and $K : n \to b$ such that $L^\dagger L = K^\dagger K$, and $b \leqslant a$ then there exist an (affine) symplectomorphism $H : b + (a - b) \to a$, such that:*



Combining lemmas 49 and 51, given two (affine) Lagrangian relations L and K with $\mathrm{im}(L^\dagger) = \mathrm{im}(K^\dagger)$, then it follows that L and K differ only up to postcomposition by an isometry! Recall the following definition from Carette et al. which quotients precisely by this congruence [Car+21, definition 7]:

Definition 52. Given a †-compact prop, C, the *discard construction* of $\mathsf{C}^{\dashv}$ is presented by freely adding a generator  $: 1 \to 0$ to C modulo the equations such that for all isometries $H : n \to m$ in C:

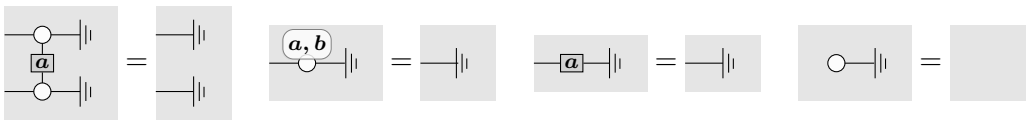


In other words, in the language of Carette et al. [Car+21, definition 10], it follows immediately from lemma 51 that $\mathsf{AffLagRel}_{\mathbb{K}}$ and $\mathsf{LagRel}_{\mathbb{K}}$ both have "enough isometries," and thus:

Theorem 53. $\mathsf{GSA}_{\mathbb{K}}^{\dashv} \cong \mathsf{AffCoIsotRel}_{\mathbb{K}}$ *and* $(\mathsf{GSA}_{\mathbb{K}}^{0})^{\dashv} \cong \mathsf{CoIsotRel}_{\mathbb{K}}$.

Luckily, the generators for (affine) symplectomorphisms, and by immediate consequence, the isometries are well-known; thus:

Corollary 54. $\mathsf{GSA}_{\mathbb{K}}^{\dashv} \cong \mathsf{AffCoIsotRel}_{\mathbb{K}}$ *is presented by the adding the generator*  $: 1 \to 0$ *to* $\mathsf{GSA}_{\mathbb{K}}$ *in addition to the equations discarding the generating isometries for all* $a, b \in \mathbb{K}$*:*



where $(\mathsf{GSA}_{\mathbb{K}}^{0})^{\dashv} \cong \mathsf{CoIsotRel}_{\mathbb{K}}$ *is presented by adding the same generator to* $\mathsf{GSA}_{\mathbb{K}}^{0}$ *and restricting the grey and white phase groups to* $\mathbb{K}$ *along the embeddings* $b \mapsto (0, b)$.

6 Applications

In mechanics, the state of a system is often given by the configurations of positions, and the evolution of the system is described by how the positions change over time. However, it is often useful to take a different perspective where states are determined by their abstract positions and momenta, ie. where they are elements of the *phase space*. From this perspective, the evolution of the system is described by how the positions and momenta change over time; in other words, the evolution is determined by the *Hamiltonian flow* of particles through the system.

6.1 Classical mechanics

In classical mechanics, the only valid reversible transformations are those which preserve the oriented local volume between position and momentum: ie the (affine) symplectormorphisms. In other words, symplectic geometry is the geometry of the phase space picture of classical mechanics. Recall from theorem 19 that a symplectic vector space can be decomposed into two parts: a Z and X grading. The elements of these gradings are interpreted respectively as abstract configurations of *positions* and *momenta*. There are different notions of positions and momenta in different classical mechanical settings (adapted from Smith [Smi93, page 23, table 2.1] and Baez [Bae13]):

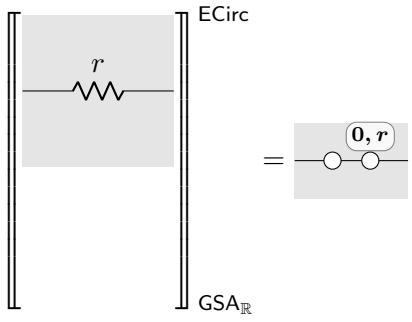
Classical mechanics	Z	dZ/dt	X	dX/dt
Translation	position	velocity	momentum	force
Electronic	charge	current	flux linkage	voltage
Hydraulic	volume	flow	pressure momentum	pressure
Thermal	entropy	entropy flow	temperature momentum	temperature

The connection between affine Lagrangian relations and electrical circuits is well documented, therefore we focus on this example. Baez et al. use linear Lagrangian relations to give a semantics for passive linear circuits: an idealized class of electrical circuits with linear behaviour [BF18]. We will review this work from a string diagrammatic perspective. To this end, recall the physical laws respectively due to Ohm [Ohm27] and Kirchhoff [Kir45]:

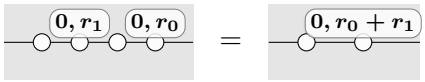
Ohm's law: The voltage around the node in a circuit is equal to the current multiplied by the resistance.

Kirchhoff's current law: The sum of currents flowing into a node is equal to the sum of currents flowing out of the node.

Given a resistor with resistance $r \in \mathbb{R}^+$, by Kirchhoff's current law the incoming current is equal to the outgoing current; and by By Ohm's law, the outgoing potential is equal to the current multiplied by r plus the incoming potential. Therefore we can interpret electrical circuits composed of resistors in $\mathsf{GSA}_{\mathbb{R}}$:

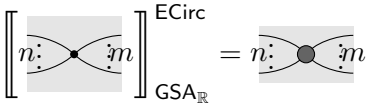


White spider fusion entails that composing resistors in sequence adds resistance:

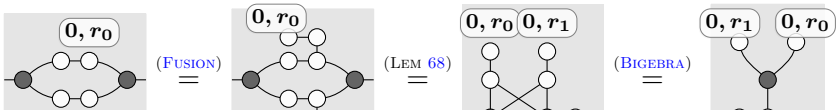


(36)

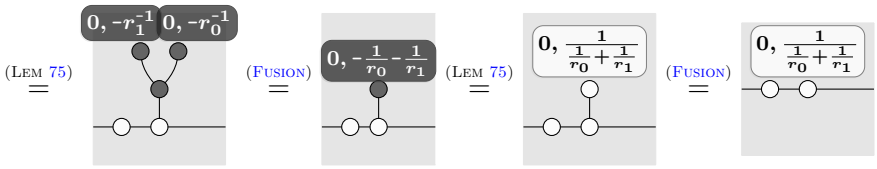
Consider an ideal junction of n incoming wires connected to m outgoing wires with no resitance. By Kirchhoff's current law, the sum of the currents of the incoming wires is equal to the sum of the currents of the outgoing wires. Moreover, by Ohm's law, because resistance is zero, the incoming and outgoing potentials are all made to be equal. Therefore an ideal junction of n incoming wires and m outgoing wires is interpreted as the following Lagrangian relation:



A circuit of resistors with resistances r_0 and r_1 composed in parallel can therefore be simplified using pivoting:

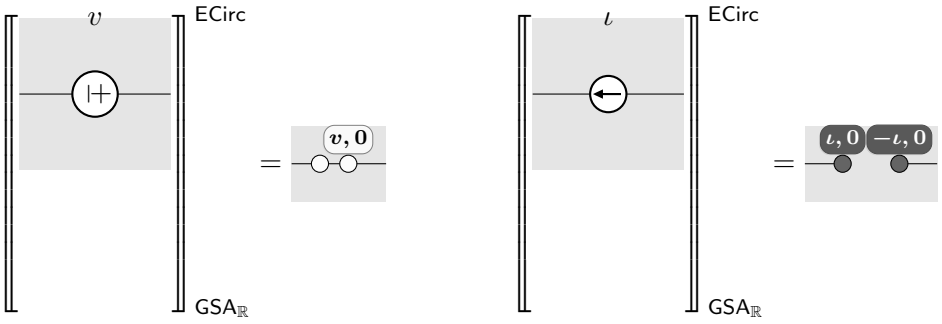


(37)

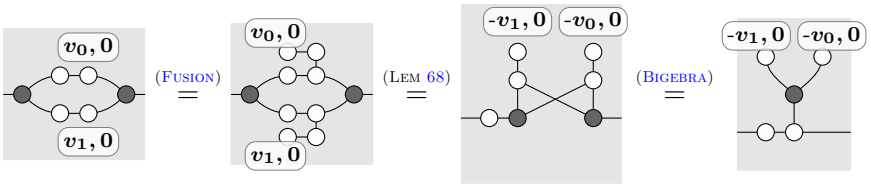


(38)

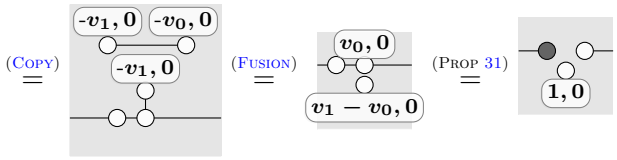
This fragment of eletrical circuits composed of circuits of resistors with junctions of wires has recently been given a complete equational theory by Cockett et al. [CKS23]; their completeness result relies on the "star-mesh" transformations, which are linear versions of the pivoting rule which we discussed in proposition 37. However, we are working in a larger semantic domain, therefore more generators can be expressed using our string diagrams. For example, by working in affine Lagrangian relations, now we can nudge the position of momentum by a constant factor. Using results from the subsequent work of Baez et al., voltage and current sources with voltage $v \in \mathbb{R}^+$ and current $\iota \in \mathbb{R}^+$ have the following interpretations [BCR18]:



We obtain an elegant proof that composing voltage sources in parallel with different voltages $v_0 \neq v_1$ leads to contradiction:

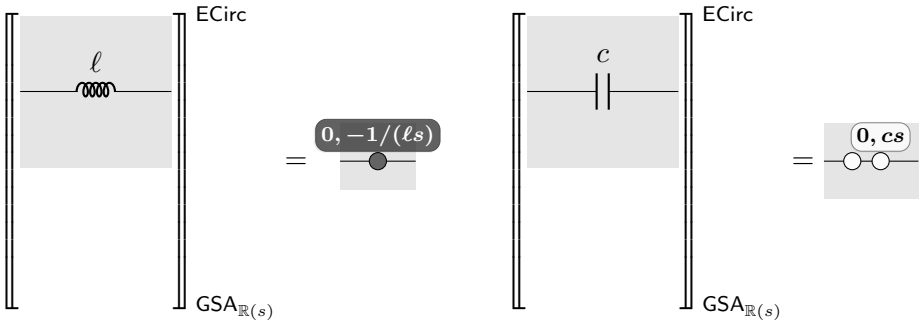


(39)

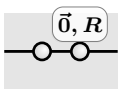


(40)

Our graphical simplifications of the parallel composition of resistors and voltage sources should be compared with the "doubled" proofs of Bonchi et al., where they represent the voltage and current gradings as separate wires in $\mathsf{GAA}_{\mathbb{R}}$ [Bon+19]. Baez et al. showed how discretized time-dependent electical circuits can be interepreted in affine Lagrangian relations over the real rational functions $\mathbb{R}(s)$ [BCR18]. In this setting, the jth element of the stream is encoded as the coefficient of the jth power of the polynomial indeterminate s. Taking the derivative and integrating with respect to time are therefore represented by multiplication and division by the polynomial indeterminate s. Applying Kirchhoff's current law and Ohm's law, (linear) inductors with inductance $\ell \in \mathbb{R}^+$ and (linear) capacitors with capacitance $c \in \mathbb{R}^+$ can therefore be interpreted as follows:



Unlike the previous approaches, the scalable notation for $\mathsf{GSA}_{\mathbb{K}}$ allows us to compactly represent electrical circuits. For example, recall that an impedance matrix is an $n \times n$ real matrix which describes the impedance between ports in an electrical network (see the textbook of Pozar [Poz11, section 4.2] for reference). An electrical network is said to be reciprocal, when the relation of impedance between nodes is symmetric: that is to say, when the corresponding impedance matrix is an element of $\mathrm{Sym}_n(\mathbb{R})$. In the scalable notation for $\mathsf{GSA}_{\mathbb{R}}$, an impedance matrix is represented by a thickened, scalable version of the resistor, which carries a symmetric matrix $R \in \mathrm{Sym}_n(\mathbb{R})$ as a phase, rather than a scalar $r \in \mathbb{R}$:



(41)

This raises the question if the constructions from Boisseau et al.'s impedance calculus [BS22] such as "impedance boxes" can be represented succinctly in the scalable notation for $\mathsf{GSA}_{\mathbb{K}}$. It is important to reiterate that the electrical circuits semantics is only one example. We could have just as easily chosen any row from the table as our semantics. For example, in fluid mechanics we would replace idealized wires with idealized pipes through which fluid flows, resistors with narrowing pipes and so on. Or more conceptually, some linear/affine network through which tokens are flowing throughout. For example, consider

an idealized network of cars driving along roads, where the flow of cars is assumed to be linear/affine. Introducing a bottleneck to the flow of traffic, such as reducing the number of lanes, would be analagous to a resistor.

6.2 Quantum mechanics

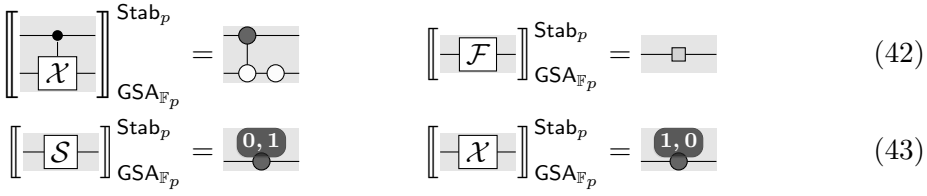
Classical mechanics can be quantized in two different, but connected ways.

6.2.1 Stabiliser quantum mechanics in odd-prime dimensions

Given an odd prime p, Comfort and Kissinger showed that the †-compact prop of affine Lagrangian relations over $\mathbb{F}_p$ is isomorphic to the †-compact prop Stab_p of odd-prime-dimensional pure qudit stabiliser circuits, modulo scalars [CK21]. If one doesn't allow symplectic-phases, then $\mathsf{GSA}_{\mathbb{F}_2}$ restricts to a semantics for qubit "CSS-circuits." Here, the Z and X gradings correspond respectively to the Pauli Z and Pauli X observables, and the symplectic form captures the commutation of Pauli operators. Therefore, we can add another row to our table:

Quantum mechanics	Z	dZ/dt	X	dX/dt
Stabiliser circuits	Pauli Z	Pauli Z flow	Pauli X	Pauli X flow

Denote the induced prop isomorphism by $[\![-]\!]^{\mathsf{Stab}_p}_{\mathsf{GSA}_{\mathbb{F}_p}} : \mathsf{Stab}_p \to \mathsf{GSA}_{\mathbb{F}_p}$. Following the convention of Gottesman [Got99], the *controlled-$\mathcal{X}$*, *Fourier*, *phase*, and *Pauli* $\mathcal{X}$ gates correspond to the generators of the prop for the affine symplectic groupoid, or "general affine symplectic group":



(42)

(43)

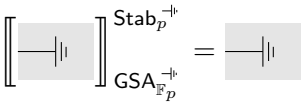
Moreover, the ground state and effect have the following interpretations:



(44)

The dagger of $\mathsf{GSA}_{\mathbb{F}_p}$ corresponds to the Hermitian adjoint in Stab_p. Therefore, we find that $\mathsf{GSA}_{\mathbb{F}_p}$ is essentially the same presentation given by Poór et al. for odd-prime-dimensional pure qudit stabiliser circuits, where we instead quotient by scalars [Poó+23]. Note that our white spiders correspond to the red spiders of Poór et al. and our grey spiders correspond to their green spiders. However, the symplectic perspective on stabiliser circuits which we take here gives a unique notational advantage over the traditional Hilbert spaces semantics which is used by Poór et al. [Poó+23] . Given the adjacency matrix G of an $\mathbb{F}_p$-edge-weighted graph on n vertices, the associated *quantum graph state* [Zho+03; Hal07; Bre15] is precisely $[\![\vec{0}, G \overset{n}{-\!\circ\!-}]\!]^{\mathsf{Stab}_p}_{\mathsf{GSA}_{\mathbb{F}_p}}$ (up to a normalisation scalar). Therefore, it is very natural to work with graph states in $\mathsf{GSA}_{\mathbb{K}}$. In our presentation one can simultanously exploit the geometric nature of string diagrams as well as the algebraic representation of adjacency matrices: here both the string diagrams are graphs, and graphs are string diagrams! Such a natural and integrated string diagrammatic representation of graph states is very important because of their widespread usage in the quantum computing literature [Kee+10; MMP13; Pro+17; Rei+19; Joo+19; Boo+23]. Comfort showed that, for odd prime p, the prop of affine coisotropic relations over $\mathbb{F}_p$ is isomorphic to odd-prime-dimensional qudit *mixed* stabiliser circuits modulo scalars, $\mathsf{Stab}_p^{+\nu}$ [Com23]. Here,

the discard relation which we added to $\mathsf{GSA}_{\mathbb{K}}$ to obtain $\mathsf{GSA}_{\mathbb{K}}^{\dashv}$ corresponds to the quantum discarding map:



(45)

The states in mixed stabiliser circuits correspond to *stabiliser codes*. Stabiliser codes are used extensively in quantum error correction, as witnessed by the success of Gottesman's PhD thesis on the subject [Got97]. Stabiliser codes are essentially stabiliser states which have been perturbed by measurement, which gives them extra redundancy to protect information against errors. Combining our work with that of Comfort [Com23], one obtains a presentation of the 2-coloured prop of stabiliser circuits with both quantum and classical types mediated by preparation and measurement maps. In fact, this has nothing to do with prime fields, and can be constructed by the coproduct of $\mathsf{GSA}_{\mathbb{K}}$ and $\mathsf{GAA}_{\mathbb{K}}$ along an isometry witnessing the embedding of †-compact props $\mathsf{GAA}_{\mathbb{K}} \hookrightarrow \mathsf{GSA}_{\mathbb{K}}$ exhibited in definition 24. In the stabiliser circuit semantics, this isometry corresponds to the "state preparation map," and its adjoint to the "measurement map." But more generally, this provides a syntax for mechanical systems with affine control.

6.2.2 Picturing quantum optics

The LOv-calculus is a complete diagrammatic language for reasoning about *passive linear* quantum optics [Clé+22]. It is a prop defined on the following set of generators, for all $\theta, \varphi \in \mathbb{R}$:

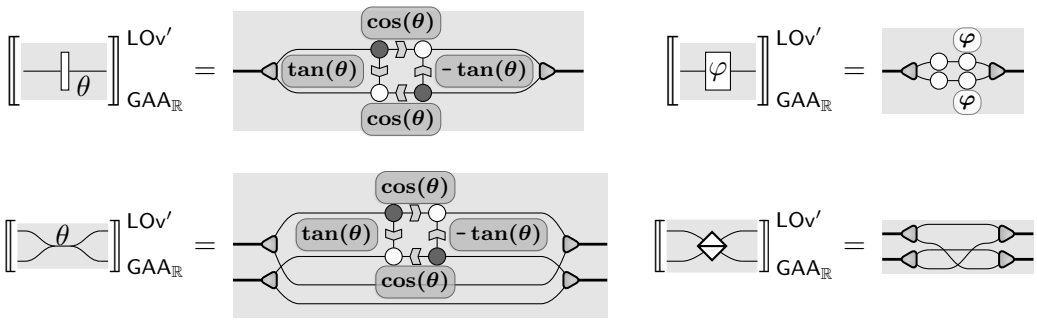
- **phase shifter**, **wave plate**, **beamsplitter**, and **polarising beamsplitter**:



- **vacuum state** and **effect**:

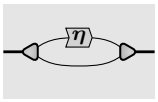


We can interpret a subset of the generators of LOv in $\mathsf{GAA}_{\mathbb{R}} \cong \mathsf{AffRel}_{\mathbb{R}}$. Let LOv' denote the fragment LOv generated only by the phase shifters, wave plates, beamsplitters and polarizing beamsplitters. Identifing the generators of LOv' with elements of the affine orthogonal group we obtain a symmetric monoidal functor $[\![-]\!]^{\mathsf{LOv}'}_{\mathsf{GAA}_{\mathbb{R}}} : \mathsf{LOv}' \to \mathsf{GAA}_{\mathbb{R}}$. On objects it is given by doubling $[\![n]\!]^{\mathsf{LOv}'}_{\mathsf{GAA}_{\mathbb{R}}} = 2n$, and on the generators by:



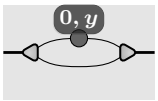
Here, there are some terminological clashes. Passive optical components correspond to *affine* relations whereas the passive electric circuits discussed in subsection 6.1 are *linear* relations. Furthermore, the phase-shifter in this context of quantum optics should therefore not be confused with the phase-shift gate in the context of stabiliser circuits which we discussed in subsubsection 6.2.1. By working in the image of the embedding $[\![n]\!]^{\mathsf{LOv}'}_{\mathsf{GAA}_{\mathbb{R}}}$ we

already have the ability to model additional quantum-optical components that are not present in LOv. For instance, active squeezing of the vertical polarisation mode is given, for $\eta \in \mathbb{R}$, by:



(46)

Furthermore $\mathsf{GAA}_{\mathbb{R}}$ is compact closed, as opposed to the usual infinite dimensional Hilbert space semantics, so by contrast we have much more freedom for rewriting in this setting. Extending along the embedding $\mathsf{GAA}_{\mathbb{R}} \to \mathsf{GSA}_{\mathbb{R}}$, we can now also describe more components of Gaussian quantum optics, since these correspond to elements of the affine *symplectic* group [Wee+12]. In this setting, shearing between the position and momentum operators takes the form:



(47)

However, one can not interpret the vacuum state in $\mathsf{AffLagRel}_{\mathbb{R}}$, because this only admits "infinitely-squeezed" states. In our forthcoming paper [BCC24], we will also show that there is a symmetric monoidal embedding $[\![-]\!]^{\mathsf{LOv}}_{\mathsf{GSA}_{\mathbb{C}}} : \mathsf{LOv} \to \mathsf{GSA}_{\mathbb{C}}$. This interpretation extends the one above and is given on the vacuum state and effect by:



(48)

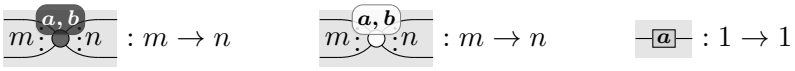
lifting the diagrammatic characterisation of quopit stabiliser quantum mechanics given in [Poó+23] to the setting of Gaussian quantum mechanics [Wee+12].

A Axiom tables

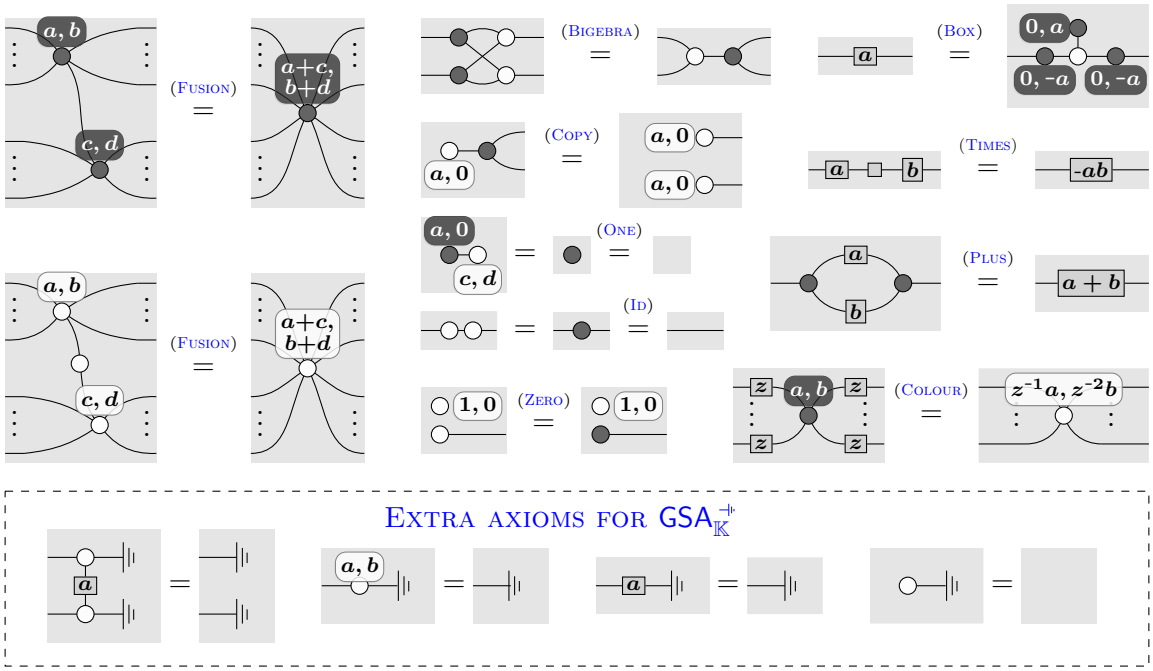
A.1 Axioms for $\mathsf{GSA}_{\mathbb{K}}$

Definition 55. Let $\mathsf{GSA}_{\mathbb{K}}$ denote the †-compact prop with:

- **Objects** natural numbers.
- **Morphisms** are presented by the following generators indexed by $n, m \in \mathbb{N}$ and $a, b \in \mathbb{K}$:



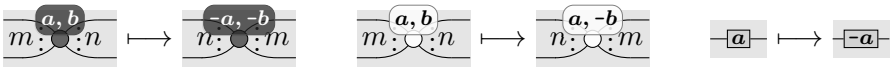
Modulo the equations imposing flexsymmetry, in addition to the following equations for all $a, b, c, d \in \mathbb{K}$ and $z \in \mathbb{K}^*$:



- **Composition** is given by horizontally connecting output wires to input wires.
- **The monoidal product** is given on objects by addition, and on morphisms by vertical juxtaposition.
- **The compact structure** is obtained from the grey Frobenius algebra:



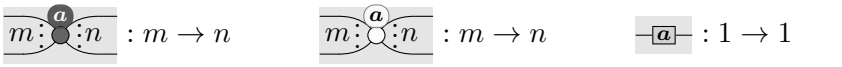
- **The dagger** is given by:



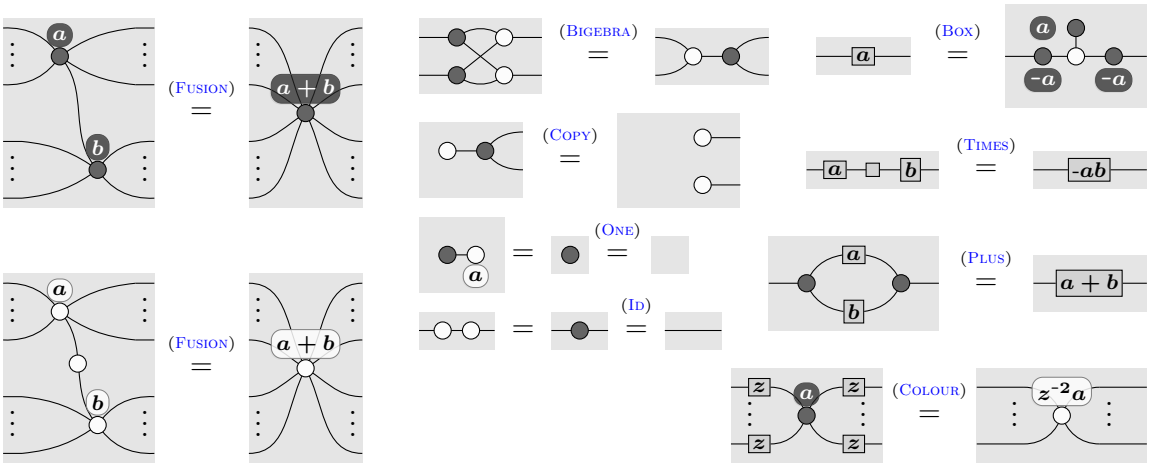
A.2 Axioms for $\mathsf{GSA}_{\mathbb{K}}^{0}$ (linear $\mathsf{GSA}_{\mathbb{K}}$)

Definition 56. Let $\mathsf{GSA}_{\mathbb{K}}^{0}$ denote the †-compact prop with:

- **Objects** natural numbers.
- **Morphisms** are presented by the following generators indexed by $n, m \in \mathbb{N}$ and $a \in \mathbb{K}$:



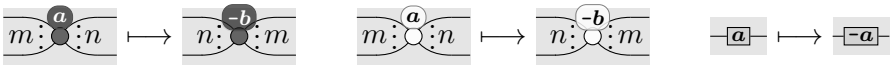
Modulo the equations imposing flexsymmetry, in addition to the following equations for all $a, b \in \mathbb{K}$ and $z \in \mathbb{K}^*$:



- **Composition** is given by horizontally connecting output wires to input wires.
- **The monoidal product** is given on objects by addition, and on morphisms by vertical juxtaposition.
- **The compact structure** is obtained from the grey Frobenius algebra:



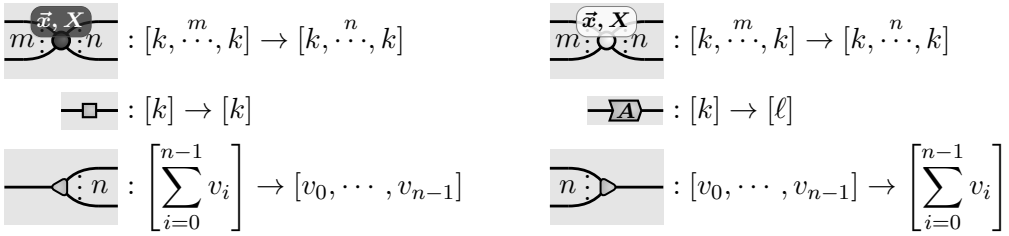
- **The dagger** is given by:



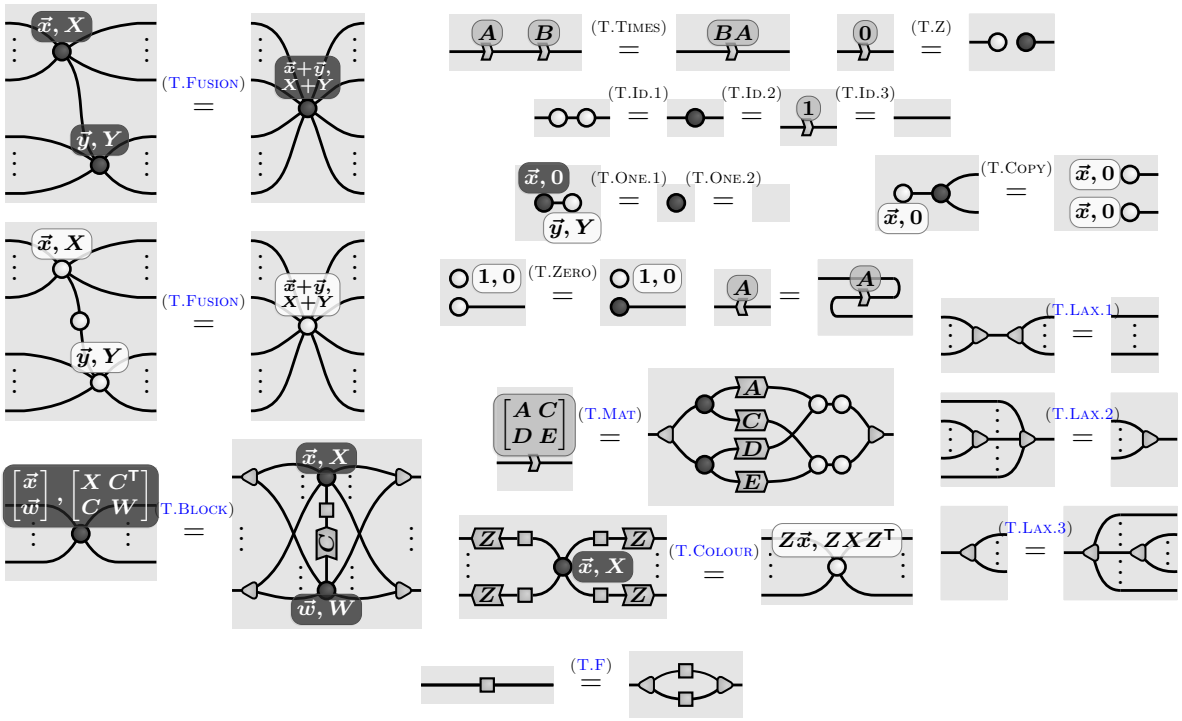
A.3 Scalable $\mathsf{GSA}_{\mathbb{K}}$

Definition 57. The strictification $[\mathsf{GSA}_{\mathbb{K}}]$ can be explicitly presented by generators and relations:

- **Objects** are elements of List($\mathbb{N}$).
- **Morphisms** are presented by the following generators indexed by $n, m, k, \ell \in \mathbb{N}$, $\vec{v} \in \mathbb{N}^n$ and $x, y \in \mathbb{K}$:



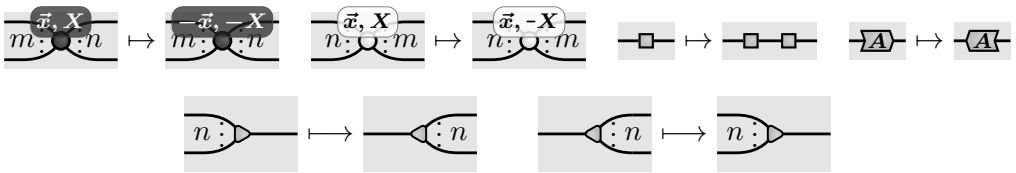
modulo the equations that the phased spiders and boxes are flexsymmetric in addition to the following equations for all $n, m, k \in \mathbb{N}$, $\vec{x}, \vec{y} \in \mathbb{K}^n$, $\vec{w} \in \mathbb{K}^m$, $X, Y \in \mathrm{Sym}_n(\mathbb{K})$, $W \in \mathrm{Sym}_m(\mathbb{K})$, $A, B, Z \in \mathrm{Mat}_{\mathbb{K}}(n, m)$, $C \in \mathrm{Mat}_{\mathbb{K}}(n, k)$, $D \in \mathrm{Mat}_{\mathbb{K}}(\ell, m)$, $E \in \mathrm{Mat}_{\mathbb{K}}(\ell, k)$, with Z surjective:



- **Composition** is given by horizontally connecting output wires to input wires.
- **The monoidal product** is given on objects by concatenation of lists, and on morphisms by vertical juxtaposition.
- **The compact structure** is obtained from the grey Frobenius algebra:



- **The dagger** is given by:



We prove (T.Fusion) in lemma 78 and (T.Colour) in lemma 83. All of the other axioms have already been proven, or follow by simple induction arguments. In the thickened setting, remark by lemma 84, we don't need a thickened version of Box. Note that we have not given a flexsymmetric presentation of the strictification of $\mathsf{GSA}_{\mathbb{K}}$ because the dividers and gathers can not be made to be symmetric!

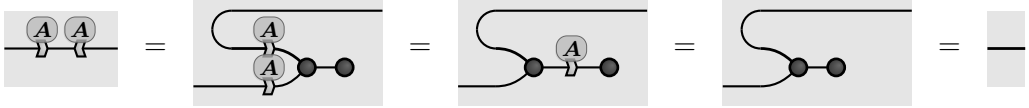
B Proofs

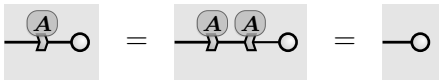
B.1 Proofs of section 3

Proof of proposition 17. We start with the first set of proposition:

(1) ⇒ (2) Taking the relational semantics, (2) can be reformulated as: for all $x, y, z \in \mathbb{K}$, $Ax = Ay = z$ is equivalent to $x = y$ and $z = Ax = Ay$. Which is true if A is injective.

(2) ⇒ (3)

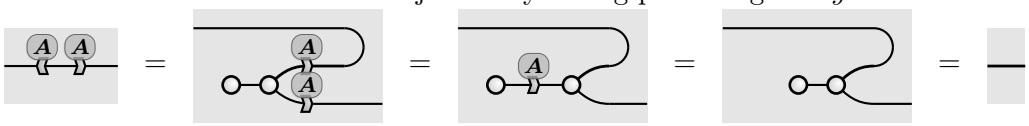


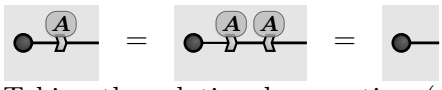
(3) ⇒ (4) 

(4) ⇒ (1) Taking the relational semantics, (4) amounts to $\ker(A) = \{0\}$ which implies that A is injective.

Then the second set of propositions:

(1) ⇒ (2) Taking the relational semantics, (2) can be reformulated as: for all $x, y, z \in \mathbb{K}$, $-Ax = y + z$ is equivalent to there exists $a, b \in \mathbb{K}$, such that $-x = a + b$, $y = Aa$ and $z = Ab$. Which is true if A is surjective by taking premimages of y and z.

(2) ⇒ (3) 

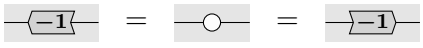
(3) ⇒ (4) 

(4) ⇒ (1) Taking the relational semantics, (4) amounts to $\mathrm{Im}(A)$ being equal to the full space, which implies that A is surjective.

□

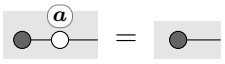
B.2 Proofs of section 4.2

Lemma 58.

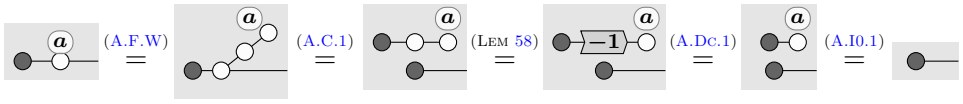


Proof. This follows immediately from (A.I1.2) and the uniqueness of inverses. □

Lemma 59.



Proof.



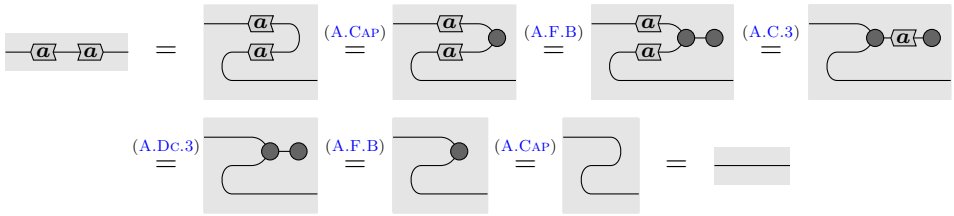
(49)

□

Lemma 60. *Given $a \in \mathbb{K}^*$:*

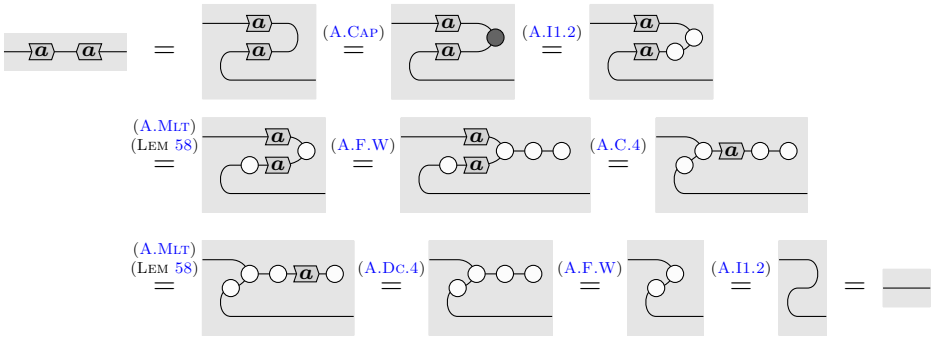


Proof. Firstly:



(50)

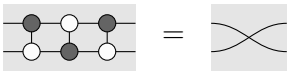
Secondly:



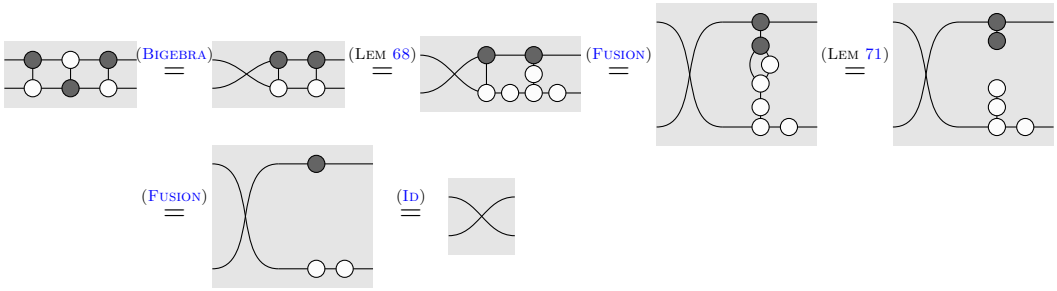
(51)

□

Lemma 61. *Given $a \in \mathbb{K}^*$:*

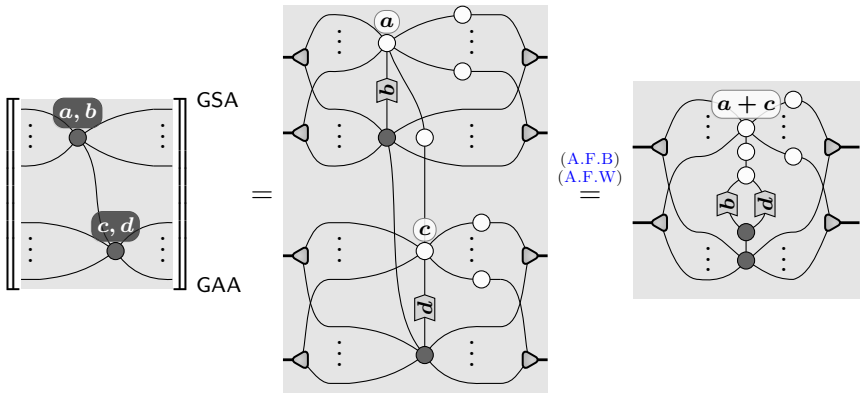


Proof.

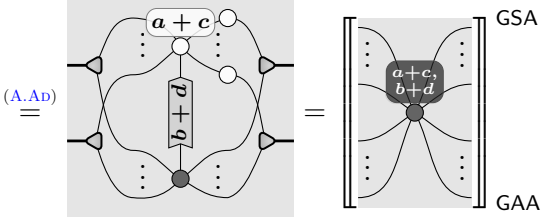


□

Proof of proposition 28. The only nontrivial thing to check is the functoriality of $[\![-]\!]^{\mathsf{GSA}}_{\mathsf{GAA}}$. To this end, we show that al of the axioms of $\mathsf{GSA}_{\mathbb{K}}$ are sound: (Fusion)

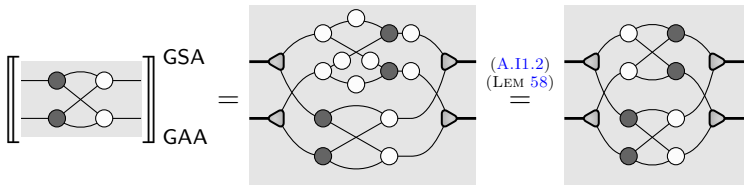


(52)

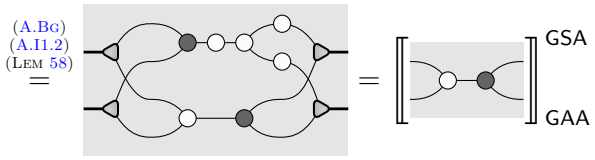


(53)

The white version of Fusion follows from an analagous argument. (Bigebra)

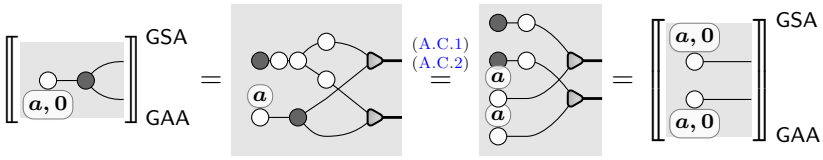


(54)



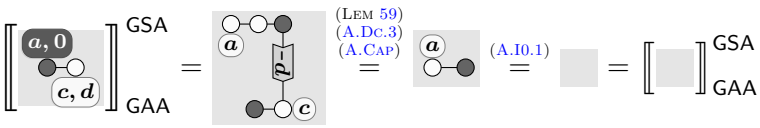
(55)

(Copy)

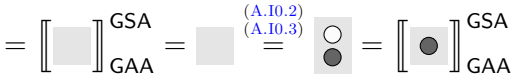


(56)

(One)

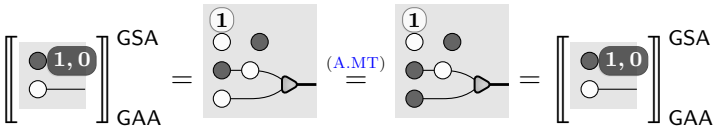


(57)



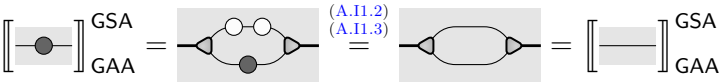
(58)

(Zero)



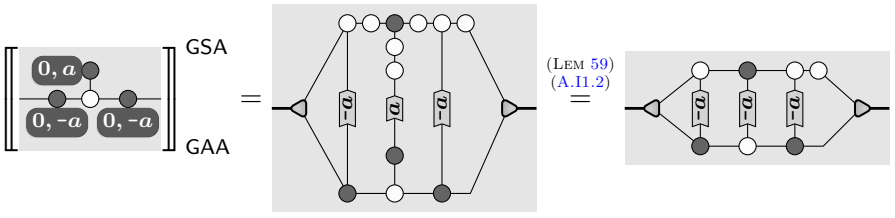
(59)

(Id)

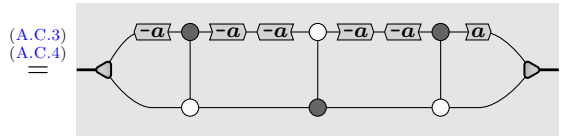


(60)

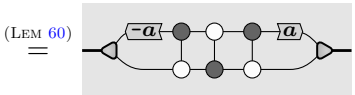
(Box) If $a \neq 0$,



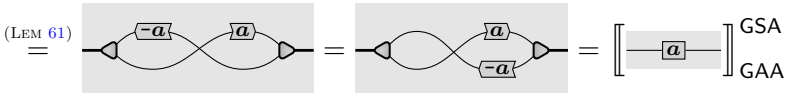
(61)



(62)

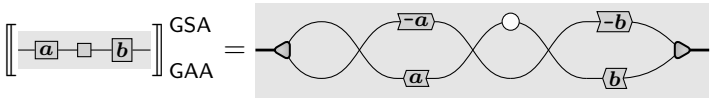


(63)

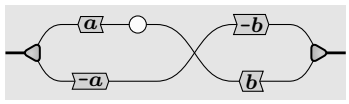


(64)

(Times) If $a, b \neq 0$,

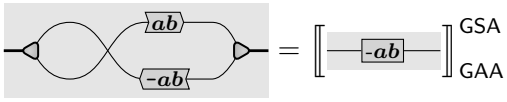


(65)



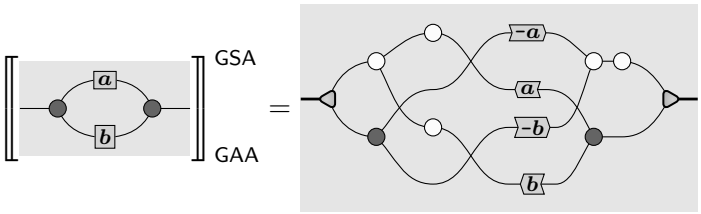
(66)

(Lem 58) (A.Mlt) =



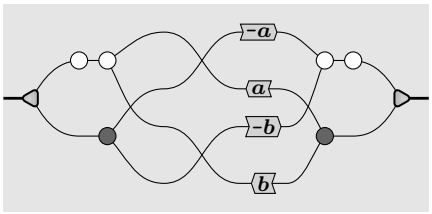
(67)

(Plus)



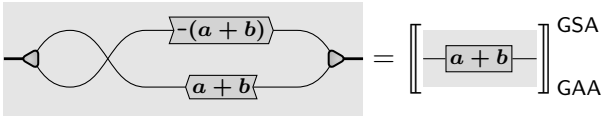
(68)

(A.F.W) (A.I1.2) =



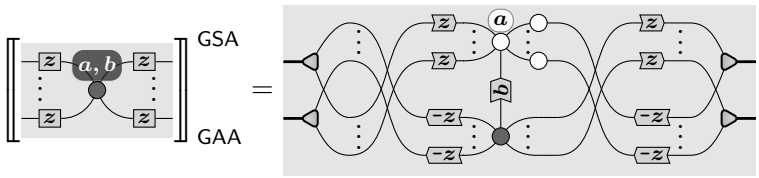
(69)

(A.Ad) =

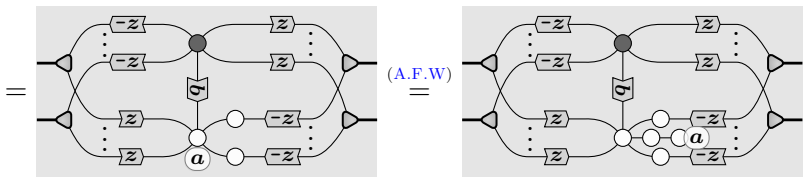


(70)

(Colour)

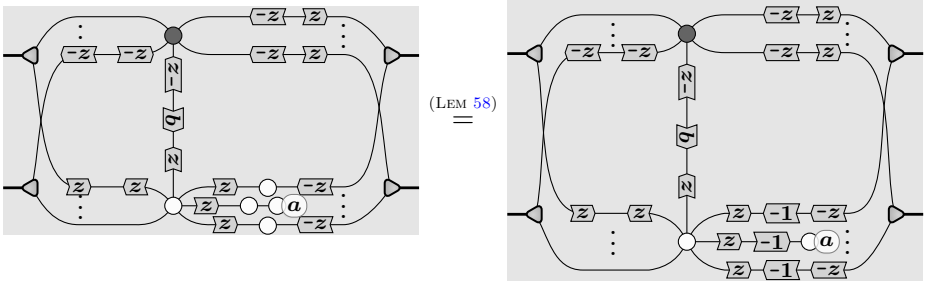


(71)

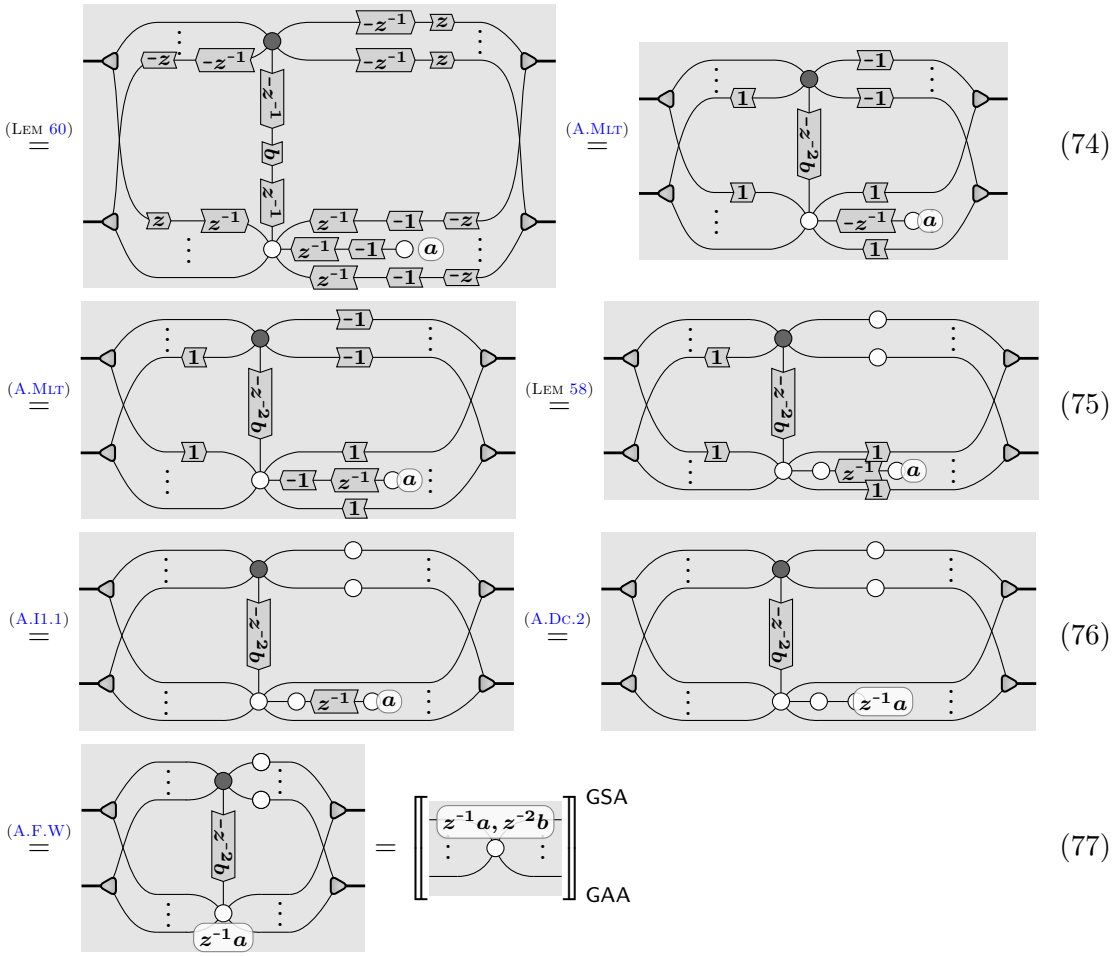


(72)

(Lem 60) (A.C.3) (A.C.4) =



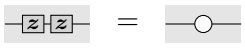
(73)



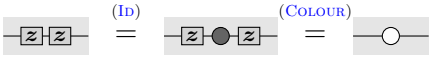
□

B.3 Proofs of section 4.4

Lemma 62. *Products of boxes are antipodes: for any $z \in \mathbb{K}^*$*

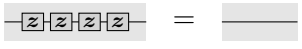


Proof.

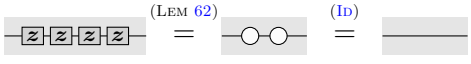


□

Lemma 63. *For any $z \in \mathbb{K}^*$*

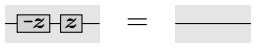


Proof.

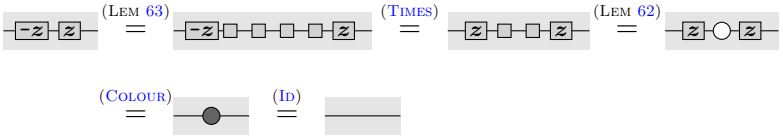


□

Lemma 64.



Proof.



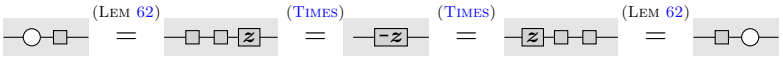
(80)

□

Lemma 65. *Boxes and antipodes commute: for any $z \in \mathbb{K}^*$,*



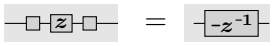
Proof.



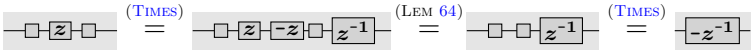
(81)

□

Lemma 66. *The opposite-inverse box can be written as follows: for any $z \in \mathbb{K}^*$,*



Proof.



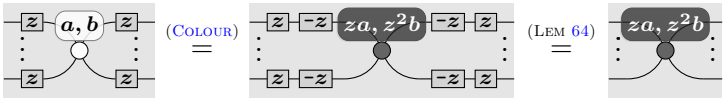
(82)

□

Lemma 67. *The "inverse" colour change rule is provable from the axioms: for any $z \in \mathbb{K}^*$,*



Proof.



(83)

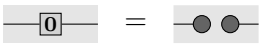
□

Lemma 68. *Spiders absorb antipodes:*

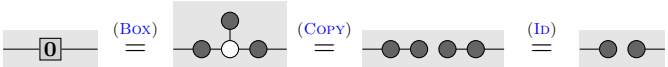


Proof. These follow straightforwardly from lemma 62, Colour and lemma 67. □

Lemma 69. *The "cut" rule for the zero box is derivable:*



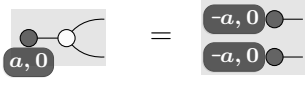
Proof.



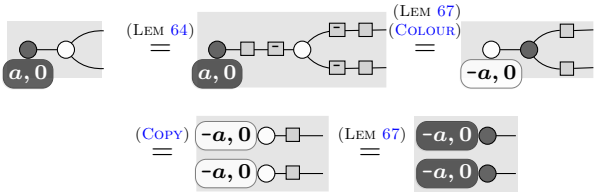
(84)

□

Lemma 70. *A colour-swapped version of Copy is derivable: for any $a \in \mathbb{K}$,*



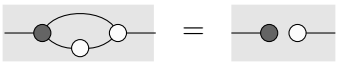
Proof.



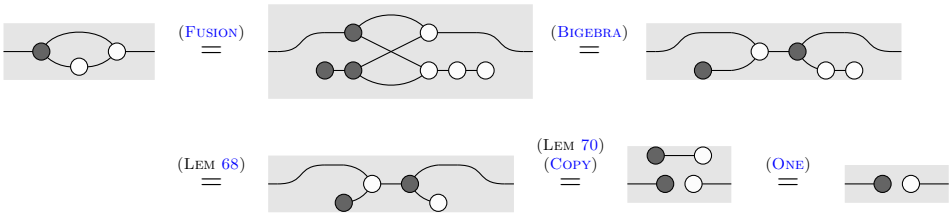
(85)

□

Lemma 71. *The Hopf identity is derivable:*



Proof.



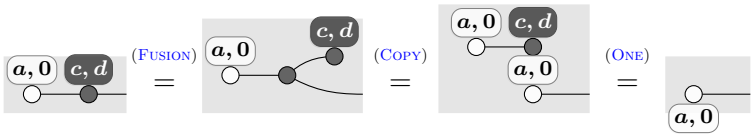
(86)

□

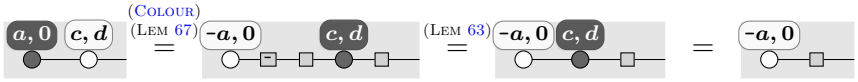
Lemma 72. *Strictly-affine white states absorb arbitrary phases and vice-versa:*



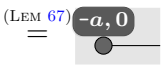
Proof.



(87)



(88)

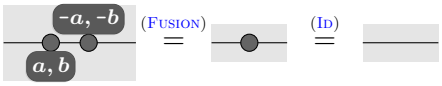


□

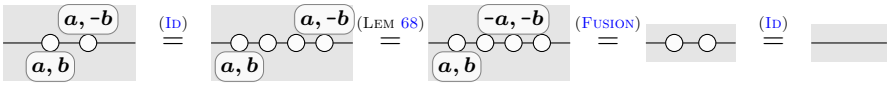
Lemma 73. *For any $a, b \in \mathbb{K}$,*



Proof.



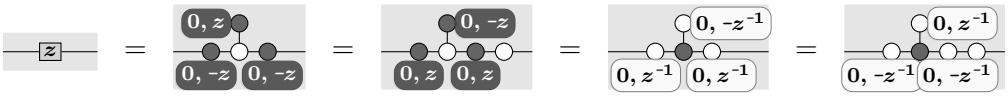
(89)



(90)

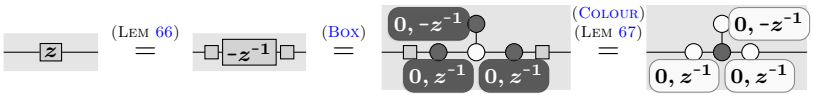
□

Lemma 74. *There are four (more or less equivalent) versions of* Box*: for any* $z \in \mathbb{K}^*$*,*

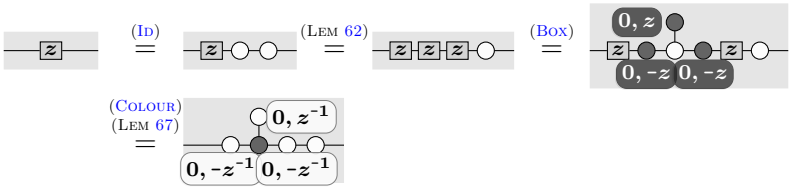


The first two equations are also valid for $z = 0$*.*

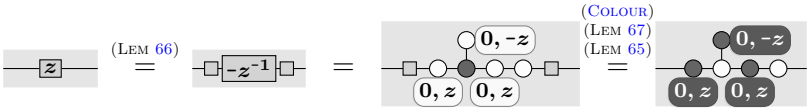
Proof. The first equation is just Box.



(91)



(92)



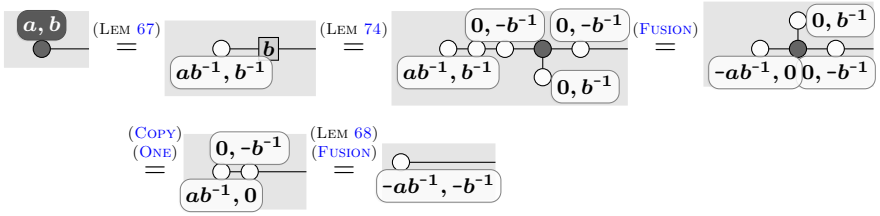
(93)

□

Lemma 75. *States with non-zero symplectic part can all be represented using both grey and white spiders: for any* $a \in \mathbb{K}$ *and* $b \in \mathbb{K}^*$*,*



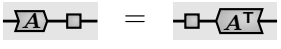
Proof.



(94)

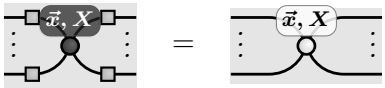
The second equation follows once again using Colour and lemma 67. □

Lemma 76. *For any* $m, n \in \mathbb{N}$ *and* $A \in \mathrm{Mat}_{\mathbb{K}}(n, m)$*,*



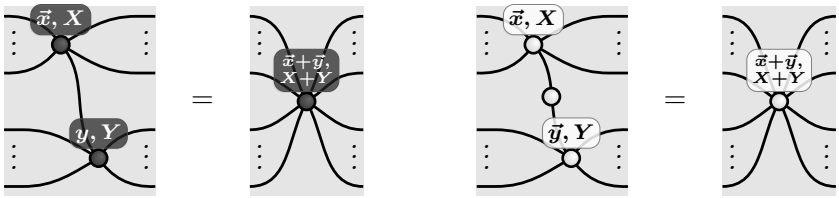
Proof. This follows from a straightforward induction on the type $m \times n$ of the matrix A. □

Lemma 77. *For any* $\vec{x} \in \mathbb{K}^n$ *and* $X \in \mathrm{Sym}_n(\mathbb{K})$*,*

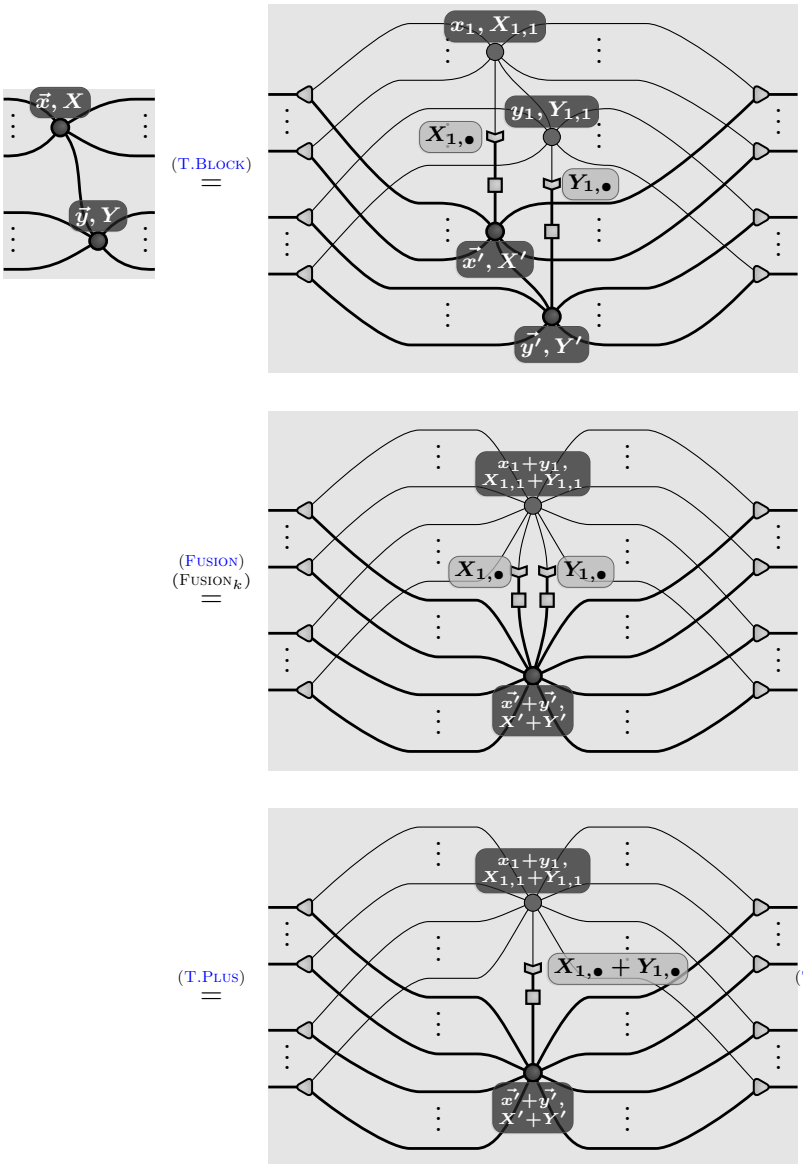


Proof. This follows straightforwardly by induction on the size n of the thick wires, using lemma 76. □

Lemma 78. *For any* $\vec{x}, \vec{y} \in \mathbb{K}^n$ *and* $X, Y \in \mathrm{Sym}_n(\mathbb{K})$,



Proof. We prove thickened grey fusion by induction on the type $n \in \mathbb{N}$ of the thick wires. The base case is trivial. Assume that the inductive claim holds for $n = k$, and denote this assumption by $\textsc{Fusion}_k$. Suppose we have two black spiders as above, of type $k+1$. Then:



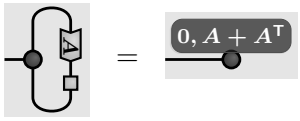
(95)

where we assume without loss of generality that

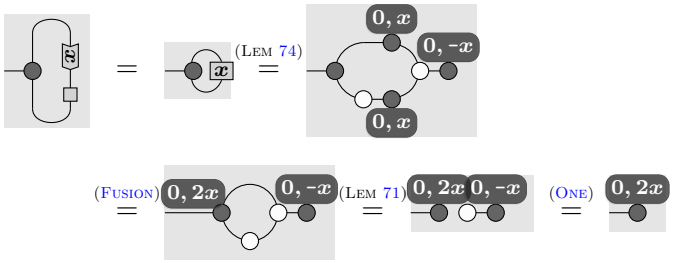
$$X = \begin{bmatrix} X_{1,1} & X_{1,\bullet}^{\mathsf{T}} \\ X_{1,\bullet} & X' \end{bmatrix} \quad \text{and} \quad Y = \begin{bmatrix} Y_{1,1} & Y_{1,\bullet}^{\mathsf{T}} \\ Y_{1,\bullet} & Y' \end{bmatrix} \tag{96}$$

The white spider fusion follows from essentially the same argument. □

Lemma 79. *For any $n \in \mathbb{N}$ and $A \in \mathrm{Mat}_{\mathbb{K}}(n,n)$,*



Proof. We proceed again by induction on the type n of the thick wires, equivalently, the dimension of the matrix A. The base case is trivial. Assume that the inductive claim holds for $n = k$, and denote this by ($\textsc{Loop}_k$), Note that for any $x \in \mathbb{K}$:

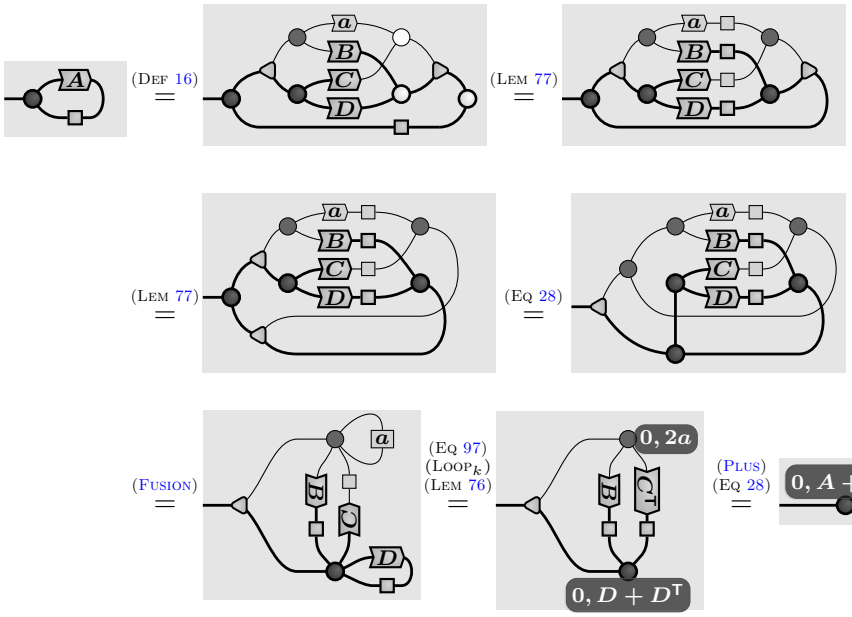


(97)

Now, let $A \in \mathrm{Mat}_{\mathbb{K}}(k+1, k+1)$, and assume without loss of generality that it takes the following form for some $a \in \mathbb{K}, B \in \mathrm{Mat}_{\mathbb{K}}(1,k), C \in \mathrm{Mat}_{\mathbb{K}}(k,1)$ and $D \in \mathrm{Mat}_{\mathbb{K}}(k,k)$:

$$A = \begin{bmatrix} a & C \\ B & D \end{bmatrix} \tag{98}$$

Then



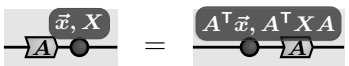
(99)

where in the last step we have identified

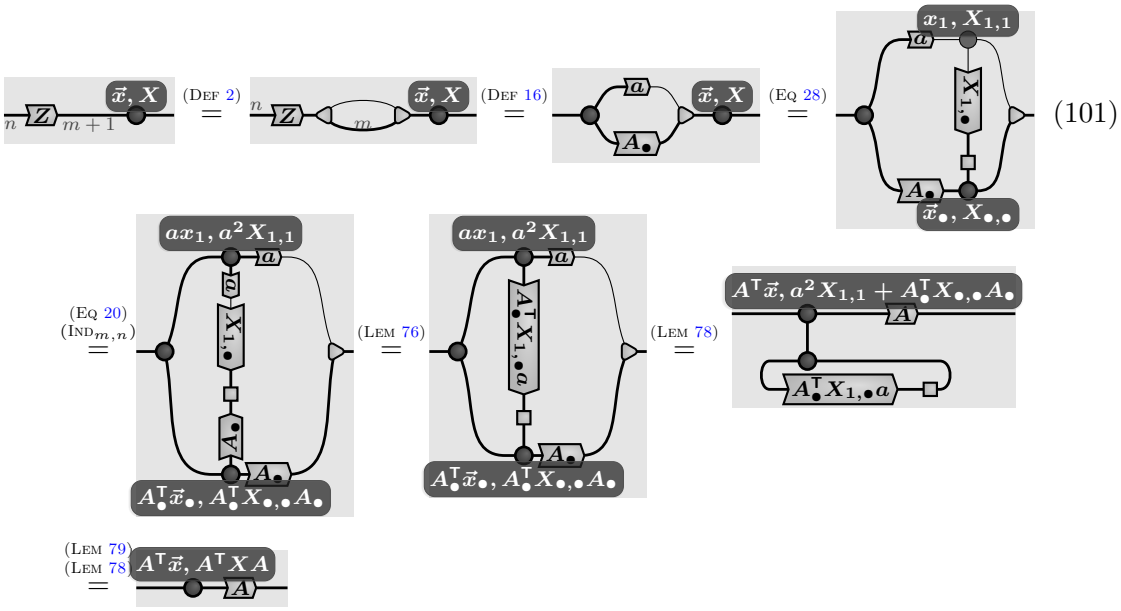
$$A + A^{\mathsf{T}} = \begin{bmatrix} 2a & C + B^{\mathsf{T}} \\ B + C^{\mathsf{T}} & D + D^{\mathsf{T}} \end{bmatrix} \tag{100}$$

□

Lemma 80. *For any $m, n \in \mathbb{N}$, any $A \in \mathrm{Mat}_{\mathbb{K}}(n,m)$, any $\vec{x} \in \mathbb{K}^m$ and any $X \in \mathrm{Sym}_m(\mathbb{K})$,*



Proof. The proof is by (double) induction on the shape $m, n \in \mathbb{N}$ of the matrix A. The base case is trivial. Now, fix m, n and assume the lemma is true for any matrix in $\mathrm{Mat}_{\mathbb{K}}(k, j)$ where either $j \leqslant m$ or $k \leqslant n$, and let $A \in \mathrm{Mat}_{\mathbb{K}}(n, m+1)$. Then, denoting the inductive step by $\textsc{Ind}_{m,n}$,

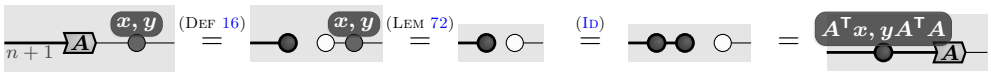


(101)

where we have identified

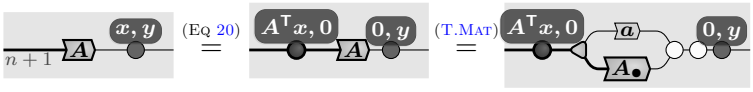
$$\vec{x} = \begin{bmatrix} x_1 \\ \vec{x}_\bullet \end{bmatrix} \qquad X = \begin{bmatrix} X_{1,1} & X_{1,\bullet}^\mathsf{T} \\ X_{1,\bullet} & X_{\bullet,\bullet} \end{bmatrix} \qquad A = \begin{bmatrix} a \\ A_\bullet \end{bmatrix} \tag{102}$$

so that $Ax = ax_1 + A_\bullet \vec{x}_\bullet$ and $A^\mathsf{T} X A = a^2 X_{1,1} + A_\bullet^\mathsf{T} X_{\bullet,\bullet} A_\bullet + a A_\bullet^\mathsf{T} X_{1,\bullet} + a X_{1,\bullet}^\mathsf{T} A_\bullet$. Secondly, fix $m \in \mathbb{N}$ and assume the lemma is true for all matrices in $\mathrm{Mat}_{\mathbb{K}}(k, j)$ such that $j \leqslant m$. Pick some $A \in \mathrm{Mat}_{\mathbb{K}}(n+1, 1)$, then if A is zero, then

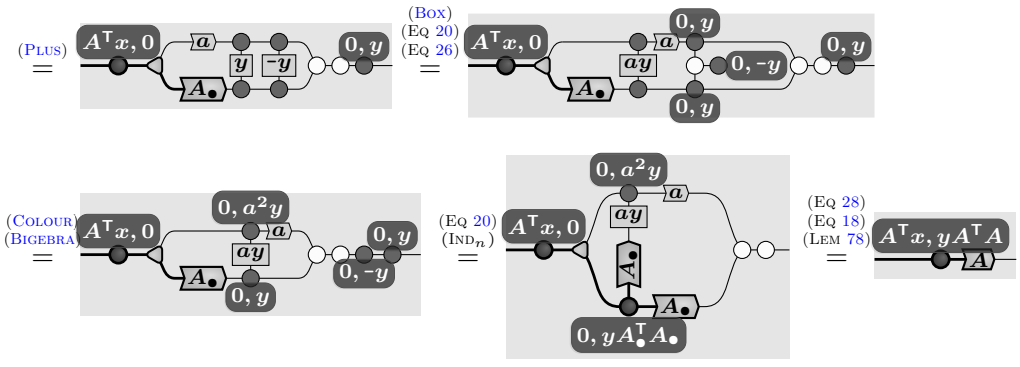


(103)

where the scalable version of Id is straightforwardly derivable by induction. Otherwise, some element of A is non-zero. For the sake of clarity, we assume that this is the first element of A, but since we can write any permutation of the elements using the swap, there is no loss of generality. Then,



(104)



where we have identified

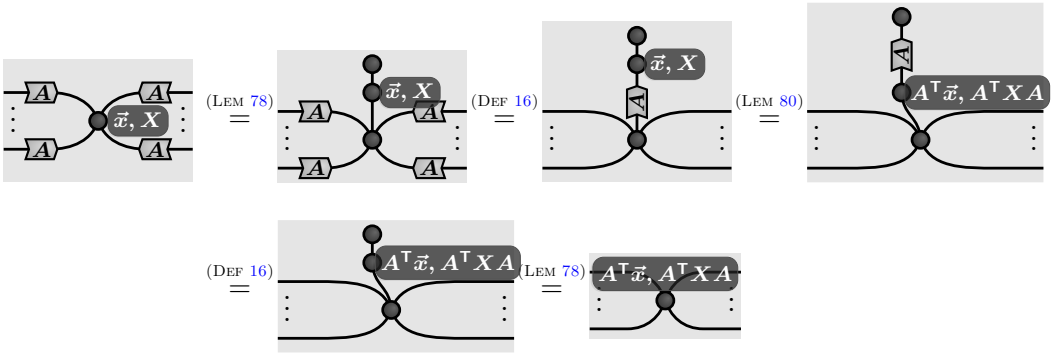
$$\begin{bmatrix} ya^2 & yaA_\bullet^\mathsf{T} \\ yaA_\bullet & yA_\bullet^\mathsf{T}A_\bullet \end{bmatrix} = yA^\mathsf{T}A \tag{105}$$

This completes the induction. □

Lemma 81. *For any $m, n \in \mathbb{N}$, any injective $A \in \mathrm{Mat}_{\mathbb{K}}(n, m)$, any $\vec{x} \in \mathbb{K}^m$ and any $X \in \mathrm{Sym}_n(\mathbb{K})$,*



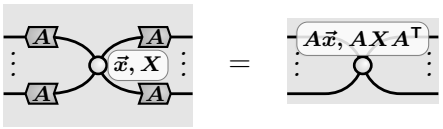
Proof.



(106)

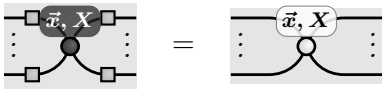
□

Lemma 82. *For any $m, n \in \mathbb{N}$, any surjective $A \in \mathrm{Mat}_{\mathbb{K}}(n, m)$, any $\vec{x} \in \mathbb{K}^n$ and any $X \in \mathrm{Sym}_n(\mathbb{K})$,*



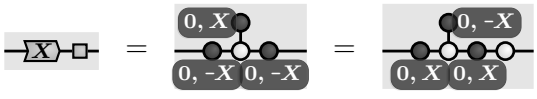
Proof. This follows from lemmas 81, 76 and 77, and the fact that the transpose of a surjective matrix is injective. □

Lemma 83.

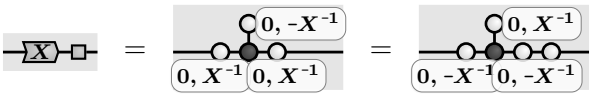


Proof. This follows immediately by combining lemmas 77 and 82. □

Lemma 84. *For any $X \in \mathrm{Sym}_n(\mathbb{K})$,*



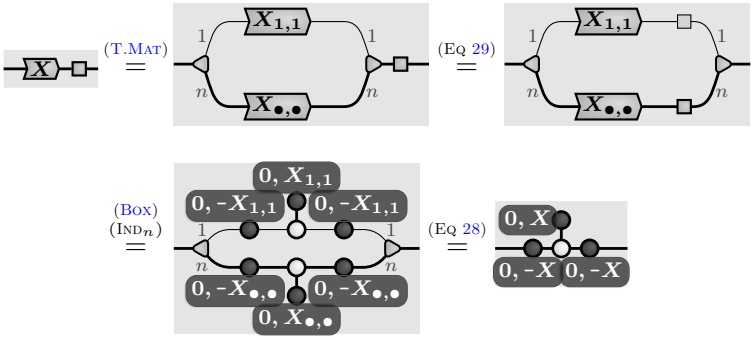
If furthermore, X is invertible,



Proof. We start by proving the first equation. The proof is by induction on the type $n \in \mathbb{N}$ of the $n \times n$ symmetric matrix X. The base case is trivial. Assume the lemma is true for all $j < n$ for some $n \in \mathbb{N}$, and pick $X \in \mathrm{Sym}_{n+1}(\mathbb{K})$, which we assume takes the form for some $X_{1,1} \in \mathbb{K}, X_{1,\bullet} \in \mathrm{Mat}_{\mathbb{K}}(1,n)$, and $X_{\bullet,\bullet} \in \mathrm{Mat}_{\mathbb{K}}(n,n)$:

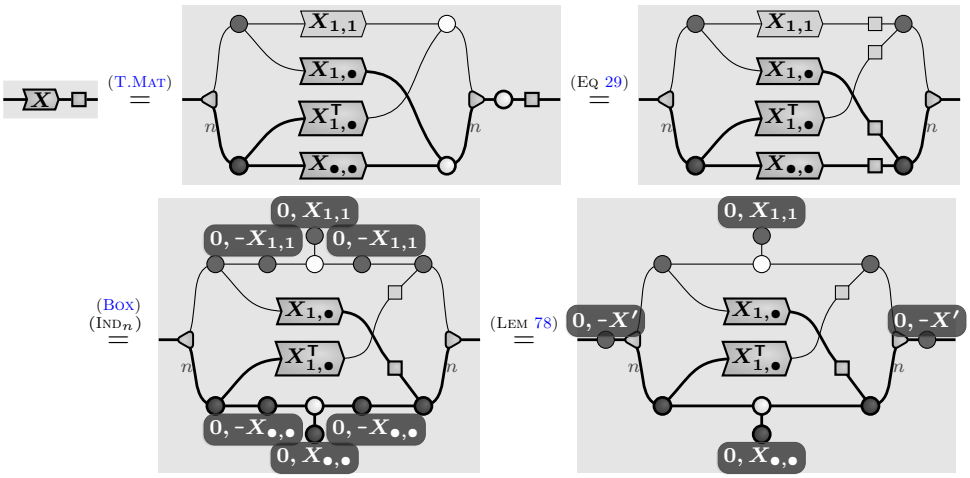
$$X = \begin{bmatrix} X_{1,1} & X_{1,\bullet}^{\mathsf{T}} \\ X_{1,\bullet} & X_{\bullet,\bullet} \end{bmatrix} \tag{107}$$

Now, if $X_{1,\bullet} = 0$, we have



(108)

Otherwise, $X_{1,\bullet}$ is injective, so that, denoting the inductive step by $\textsc{Ind}_m$,

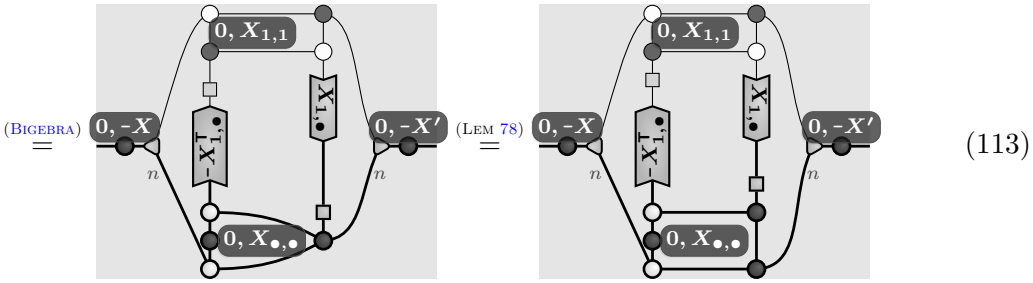


(109)

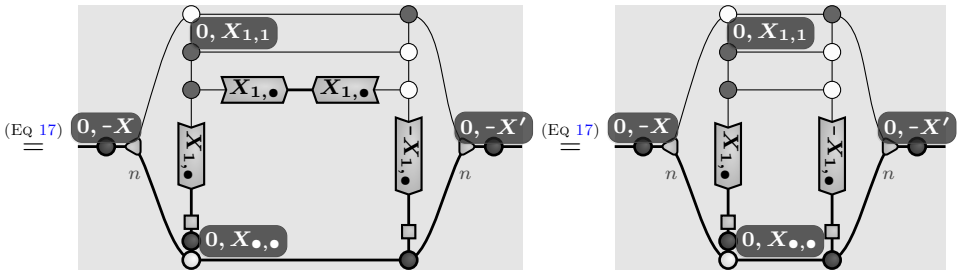
(110)

(111)

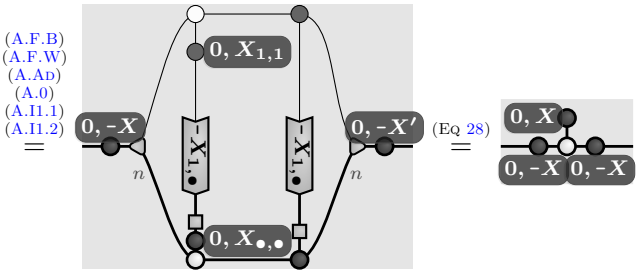
(112)



(113)



(114)



(115)

This completes the induction. The remaining equations follow from manipulations analogous to the proof of lemma 74. □

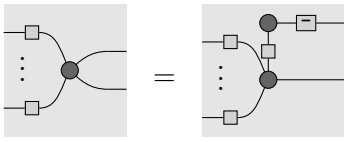
Lemma 85. *For any $n \in \mathbb{N}$, $\vec{x} \in \mathbb{K}^n$ and invertible $X \in \mathrm{Sym}_n(\mathbb{K})$,*



Proof. The proof is analogous to the proof of lemma 75. □

Proof of proposition 34. Transform the diagram as follows:

1. Use Colour to change every white spider into a grey one, surrounded by boxes. The resulting diagram consists of only grey spiders and boxes.
2. Use Id to add a grey spider between any two subsequent boxes. The diagram now consists of only grey spiders, connected either by plain edges or boxes.
3. Use Fusion to fuse any two spiders connected by plain edges, to eliminate any loops, and use lemma 79 to eliminate any "loops" carrying boxes. The resulting diagram contains no loops, and furthermore spiders are connected only by edges carring boxes.
4. Use Plus to fuse all edges between two spiders into a single weighted box or no edge (after a simple application of lemma 69 and Fusion).
5. Use Id and lemma 64 to obtain

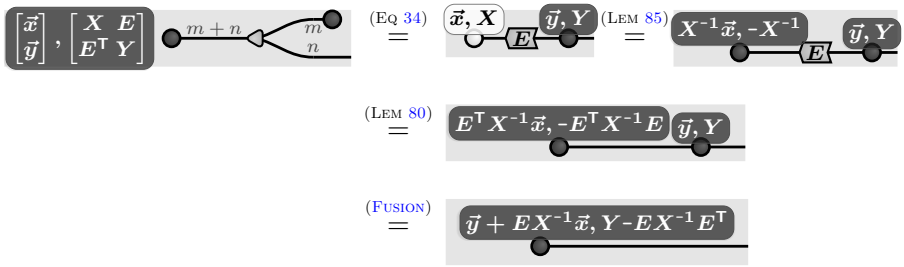


(116)

which allows one to split any spider connected to more than one output into several grey spiders, each connected to exactly one output.

The resulting diagram is in graph-like form. □

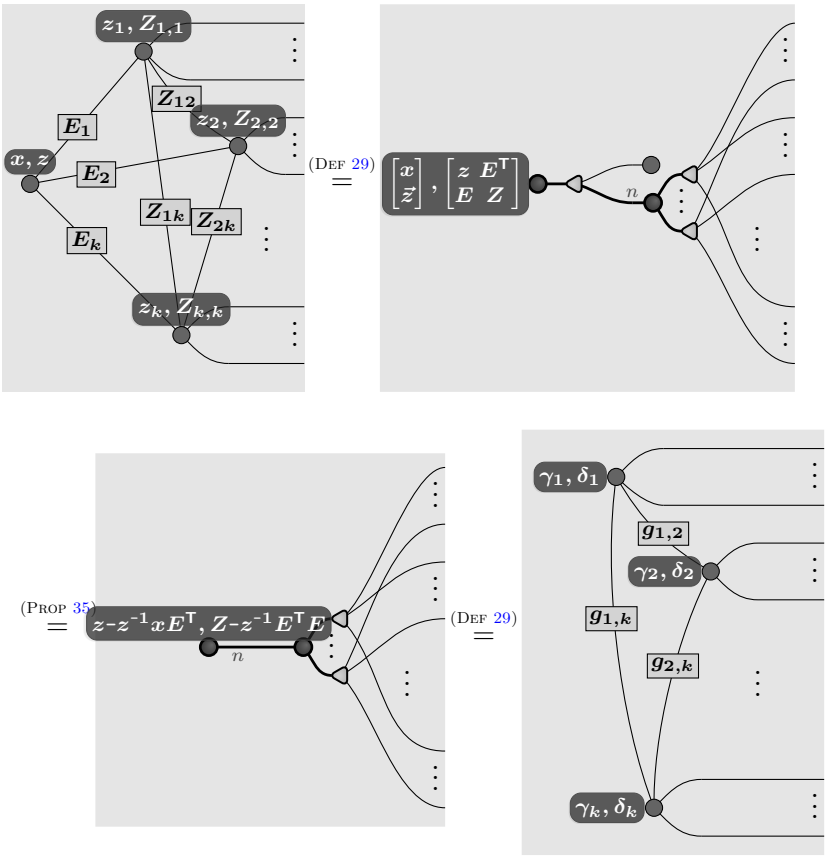
Proof of proposition 35.



(117)

□

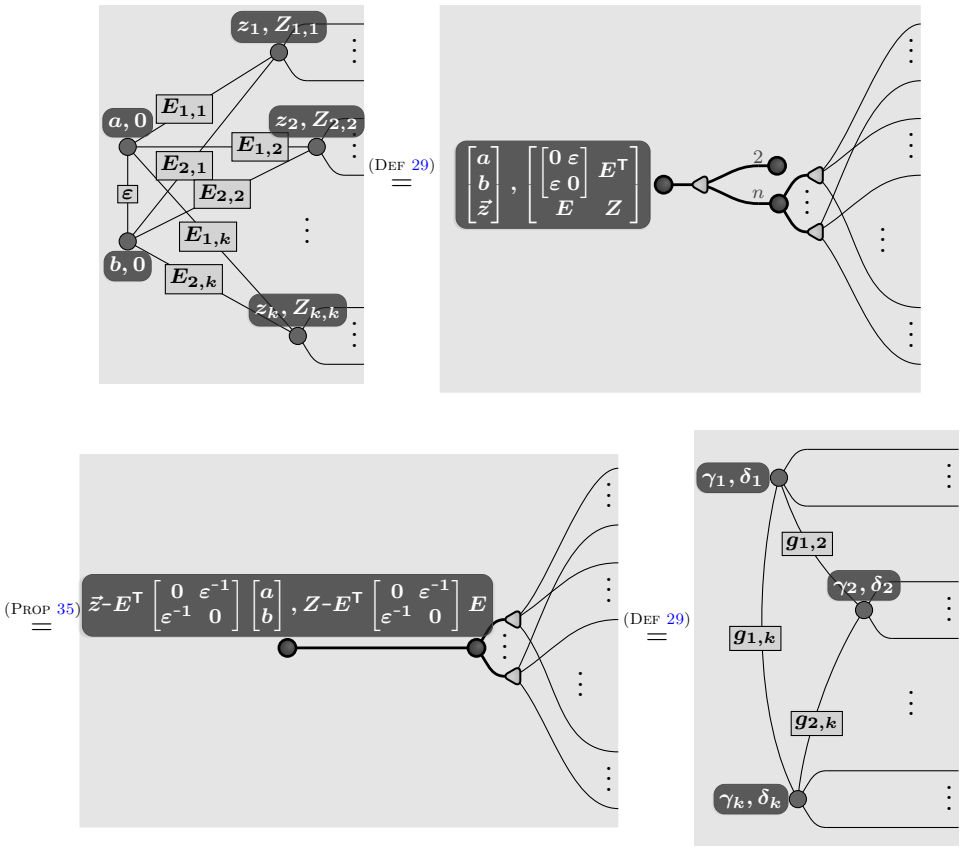
Proof of proposition 36.



(118)

where $E \in \mathrm{Mat}_{\mathbb{K}}(1,k)$ has components E_j and $Z \in \mathrm{Sym}_k(\mathbb{K})$ has components $Z_{j,k}$. In the last diagram, we have $\gamma_i = z_i + E_i x z^{-1}$, $\delta_i = Z_{i,i} - z^{-1}E_i^2$, and $g_{i,j} = Z_{ij} - z^{-1}E_iE_j$ as required. □

Proof of proposition 37.



(119)

where we have applied proposiiton 35 to the invertible submatrix

$$\begin{bmatrix} 0 & \varepsilon \\ \varepsilon & 0 \end{bmatrix} \quad \text{whose inverse is} \quad \begin{bmatrix} 0 & \varepsilon^{-1} \\ \varepsilon^{-1} & 0 \end{bmatrix}. \tag{120}$$

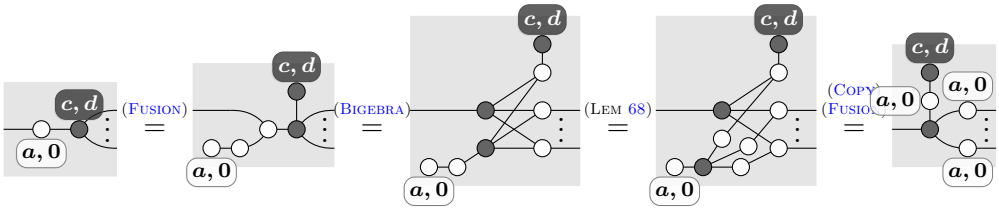
The matrix $E \in \mathrm{Mat}_{\mathbb{K}}(2,k)$ has components $E_{j,k}$ and $Z \in \mathrm{Sym}_k(\mathbb{K})$ has components $Z_{j,k}$. Here, we have $\gamma_i = z_i + \varepsilon^{-1}(aE_{2,i} + bE_{1,i})$, $\delta_i = Z_{i,i} - 2\varepsilon^{-1}E_{1,i}E_{2,i}$, and $g_{i,j} = -\varepsilon^{-1}(E_{1,i}E_{2,j} + E_{1,j}E_{2,i})$ as requied. □

Proof of proposition 39. Given a graph-like diagram, keep appling propositions 36 and 37 to the internal vertices of the diagram as long as it is possible. Every application strictly reduces the number of internal vertices hence the algorithm must halt. The result is a diagram in AP-form since proposition 36 eliminates all internal vertices with non-zero symplectic phase, and proposition 37 eliminates all pairs of connected internal vertices. □

Lemma 86. *White spiders with arbitrary phases copy affine grey spiders and vice-versa:*

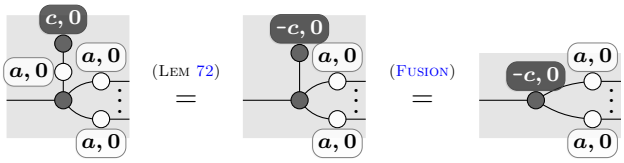


Proof. First of all,



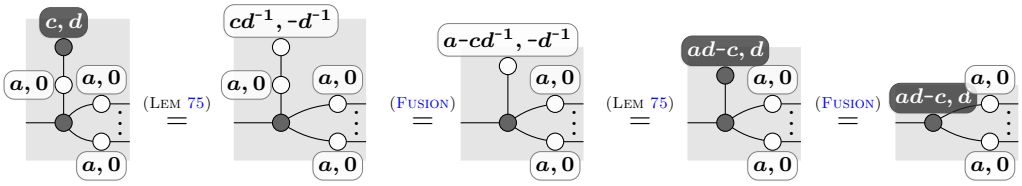
(121)

Then, we separate the equation into two cases based on whether the grey spider is affine or not. In case $d = 0$, the grey spider is affine and therefore:



(122)

Note that if $d = 0$, then $ad - c = -c$ and so the lemma holds. Otherwise, $d \neq 0$ and therefore d^{-1} exists, so we can apply the state-change lemma:

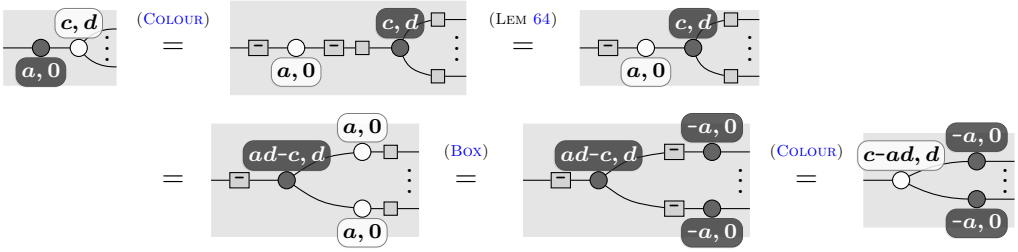


(123)

Note that the phases after the application of the second state-change follow from:

$$-(a - cd^{-1})(-d^{-1})^{-1},\ -(-d^{-1})^{-1} = -(a - cd^{-1})(-d),\ d = ad - c,\ d \tag{124}$$

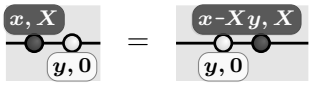
We can prove the second equation of the lemma using Hadamard-boxes as follows:



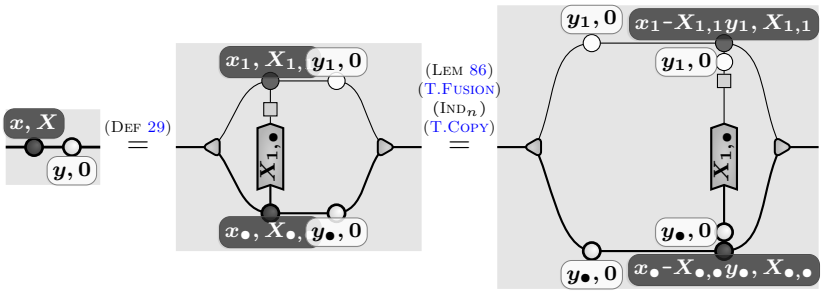
(125)

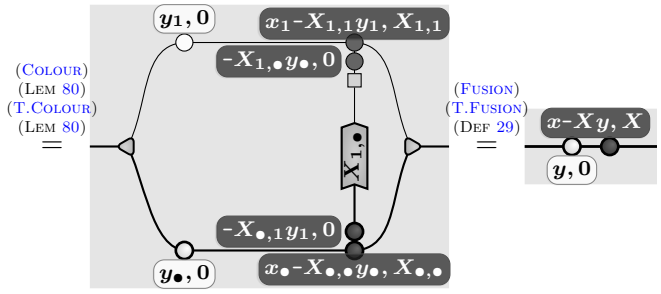
□

Lemma 87. *For any $x, y \in \mathbb{K}^n$ and $X \in \mathrm{Sym}_m(\mathbb{K})$,*



Proof. We prove the claim by induction on the thickness of scalable spiders. The base case is trivial. For the inductive step:

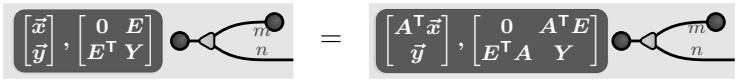




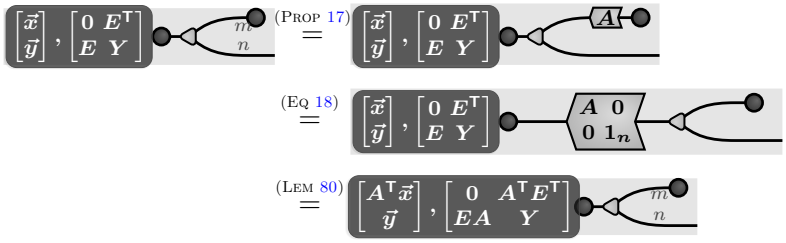
□

We can perform row operations on the matrix E of a $\mathsf{GSA}_{\mathbb{K}}$-diagram in AP-form without changing the semantics. However, applying these row operations can change the affine phases $\vec{x}$ of the internal spiders:

Lemma 88. *Given a diagram in AP-form and an* invertible *matrix* $A \in \mathrm{Mat}_{\mathbb{K}}(m,m)$,



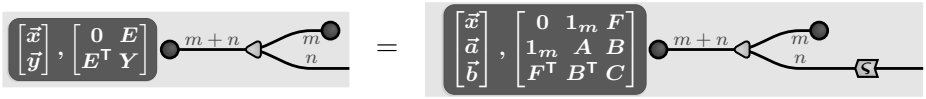
Proof.



(127)

□

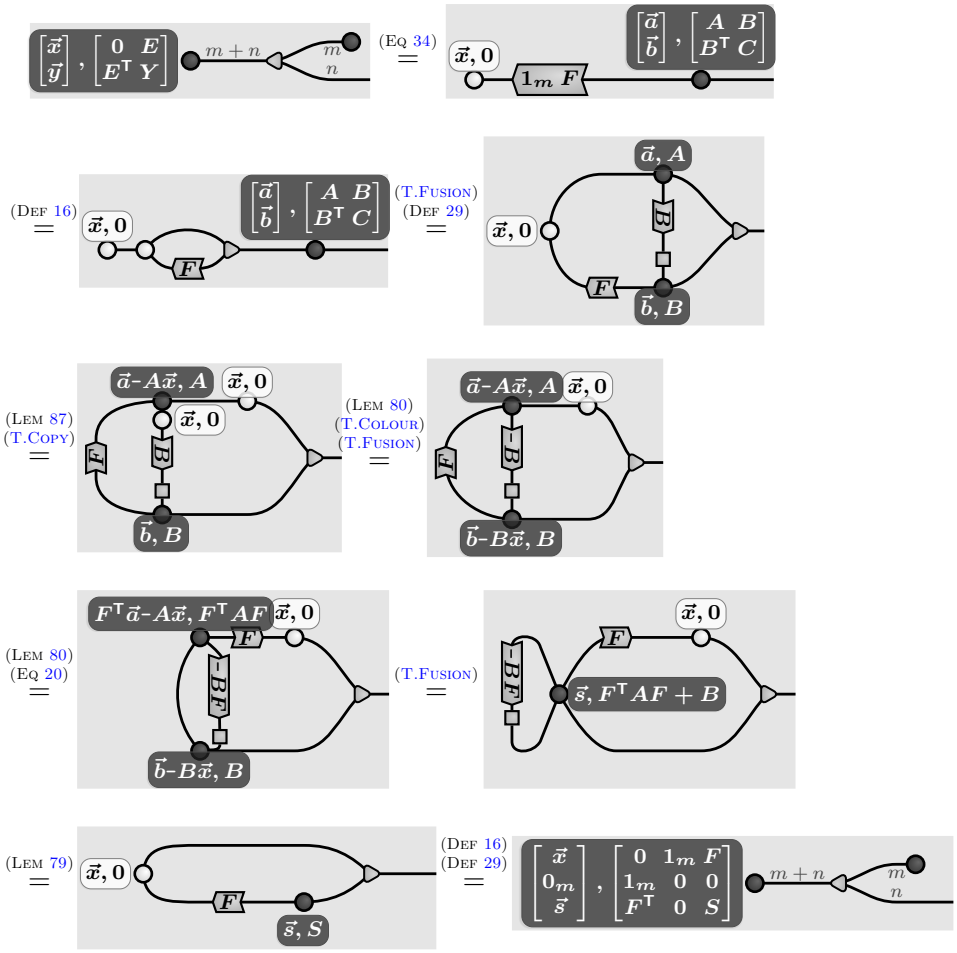
Proof of proposition 43. Consider a diagram in AP-form described by the quadruplet $(E, Y, \vec{x}, \vec{y})$. Then, by lemma 88 we can act on E on the left by an invertible matrix A. Thus, we can run a Gaussian elimination on E. During this procedure, if at any point the resulting matrix AE has a zero row, this means that one of the internal vertices has no neighbours in the graph. This disconnected internal vertex then takes the form . If $x \neq 0$ then the diagram has empty semantics and we can use lemma 31 to rewrite the diagram to normal form. Otherwise, $x = 0$ and we can eliminate the disconnected vertex using One. The resulting row reduced echelon form of E, with zero rows removed, is of the form $\begin{bmatrix} 1_m & F \end{bmatrix}$, up to a permutation of its columns. Denoting this permutation by ς, we have



(128)

for matrices A, B, C, F and vectors $\vec{a}, \vec{b}$ of suitable dimensions. Let's ignore the permuta-

tion for now, then following equation (34) we have



(129)

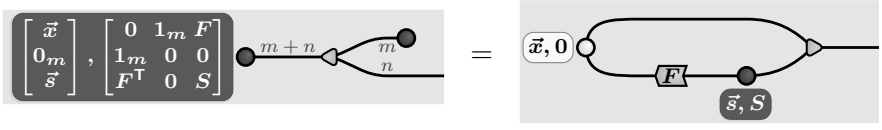
where we have set $\vec{s} := F^\mathsf{T}\vec{a}+\vec{b}-(A+B)\vec{x}$ and $S := F^\mathsf{T}AF+B-BF-F^\mathsf{T}B^\mathsf{T}$. Reintroducing the permutation, we see that this final diagram is exactly the form described in remark 42, which is easily seen to verify the conditions of definition 41. The diagram is therefore in reduced AP-form. □

Lemma 89. *The thick scalable version of lemma 72 holds:*



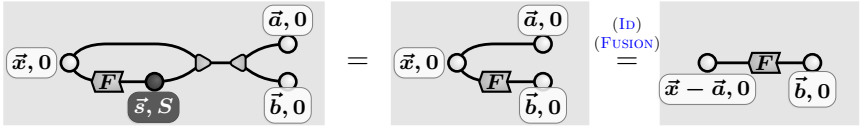
Proof. The proof is essentially the same as that of lemma 72 □

Proof of proposition 44. Up to permutation the proof of proposition 43, entails that the the AP-form can be rewritten to:



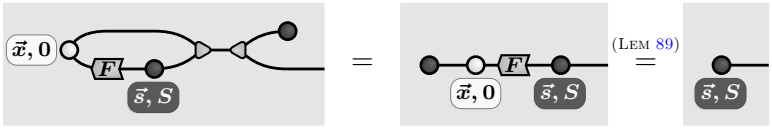
(130)

Note that the number m of internal vertices is an invariant of the reduced AP-form, as is the permutation of the output wires ς. The affine subspace $(F, \vec{x})$ is also invariant as:



(131)

As is the affine subspace $(S, \vec{s})$:

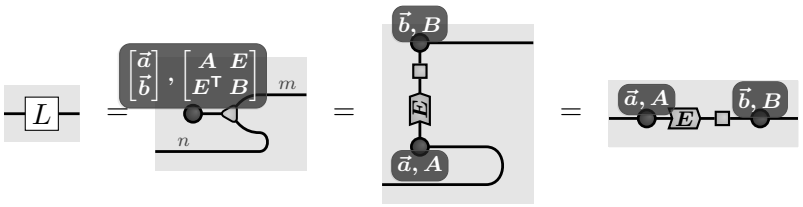


(132)

□

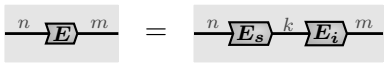
B.4 Proofs of section 5

Proof of Lemma 50. Given a diagram $L : n \to m$ in $\mathsf{GSA}_{\mathbb{K}}$, by proposition 34, it can be rewritten into graph-like form. Moreover, by propositions 36 and 37, we can remove all of the internal edges to obtain a diagram of the following form, for some $a \in \mathbb{K}^n$, $A \in \mathrm{Sym}_n(n)$, $b \in \mathbb{K}^m$, $B \in \mathrm{Sym}_m(m)$, and $E \in \mathrm{Mat}_{\mathbb{K}}(n, m)$:



(133)

Moreover, any such matrix E can be factorized into a surjection $E_s \in \mathrm{Mat}_{\mathbb{K}}(n, k)$ followed by an injection $E_i \in \mathrm{Mat}_{\mathbb{K}}(k, m)$ where $k = |\operatorname{im}(E)|$:



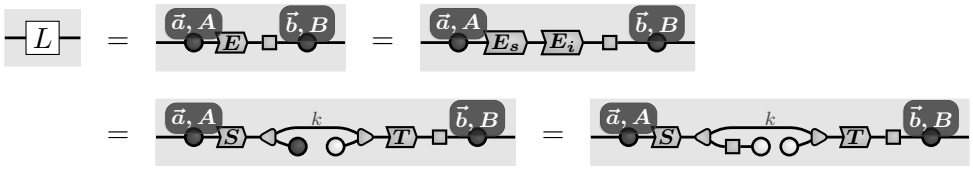
(134)

Where the surjection E_s can moreover decomposed into an isomorphism $S \in \mathrm{Mat}_{\mathbb{K}}(n, n)$ followed by a canonical projection onto k; and similarly, the injection E_i can be decomposed into a canonical injection into k followed by an isomorphism $T \in \mathrm{Mat}_{\mathbb{K}}(m, m)$:



(135)

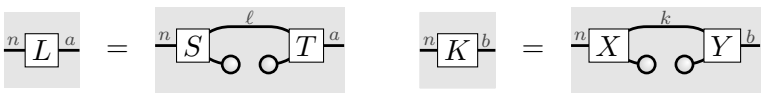
Therefore:



(136)

Where we remark that $\operatorname{im}(L) \cong \operatorname{im}(E)$, thus $|\operatorname{im}(L)| = |\operatorname{im}(E)| = k$, as desired. □

Proof of lemma 51. Take two arrows $L : n \to a$ and $K : n \to b$ in $\mathsf{GSA}_{\mathbb{K}}$ such that $L^\dagger L = K^\dagger K$ and $a \leqslant b$. By lemma 50, there are $\ell, k \in \mathbb{N}$ and affine symplectomorphisms $S, X : n \to n$, $T : a \to a$, and $T : b \to b$ such that:



(137)

Because $L^\dagger L = K^\dagger K$, by lemma 48:

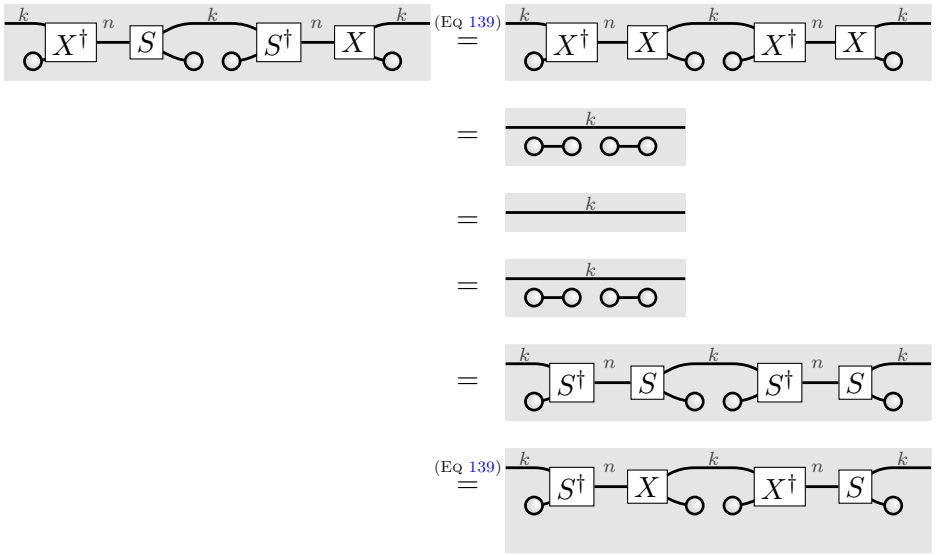
$$\ell = \left|\operatorname{im}(L^\dagger)\right| = \left|\operatorname{im}(K^\dagger)\right| = k \tag{138}$$

Moreover: because $L^\dagger L = K^\dagger K$, we have:



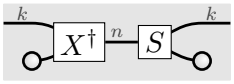
(139)

Furthermore, remark that:



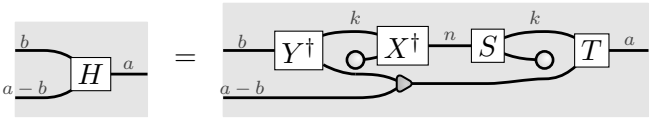
(140)

Therefore, the following map is unitary (equivalently, it is an affine symplectomorphism):



(141)

And thus, the following map is unitary as well:



(142)

Therefore:



(143)

□

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